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(54) **FORMWORK FOR RETAINING WALLS**

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(57) **ABSTRACT**

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A waler adapted for use as part of a modular cast concrete retaining wall system; said waler being adapted to interconnect with adjacent walers; said waler having side walls and upper walls defining an internal casting cavity; wherein said waler is adapted to be oriented in use in a substantially horizontal alignment by having relatively concrete impenetrable side walls and apertures in said upper and lower walls, thereby to permit passage of fluid concrete into and through said casting cavity; and a system incorporating same.

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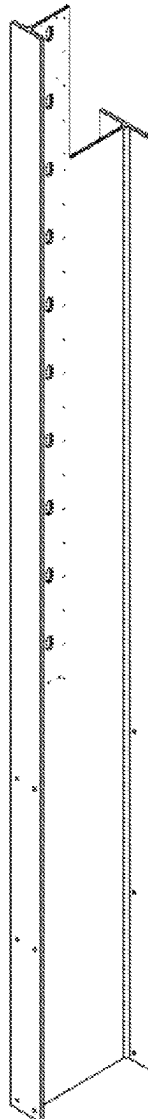


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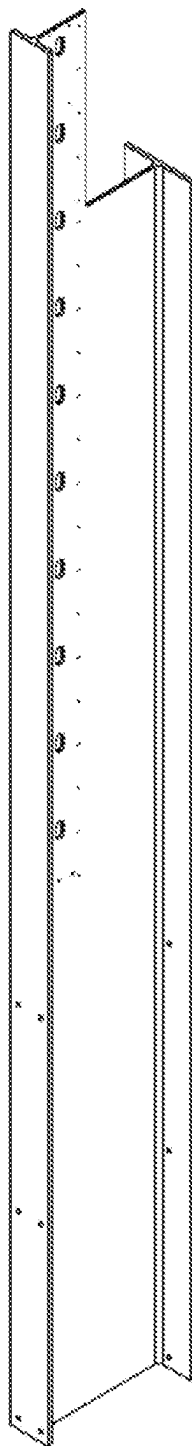


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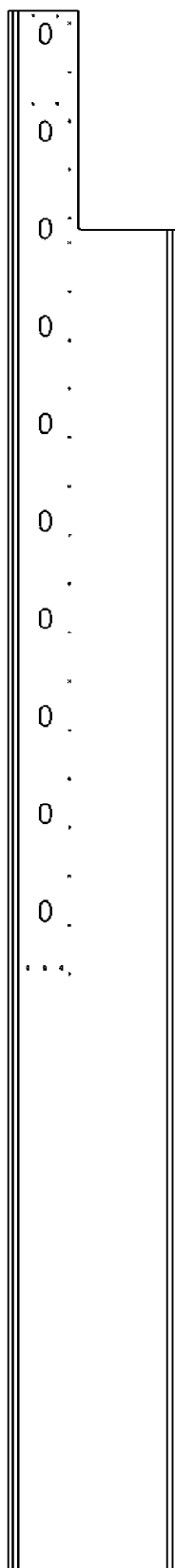


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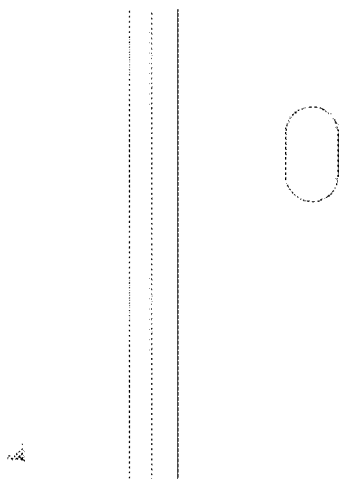


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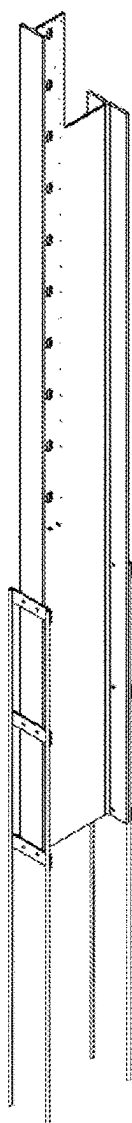


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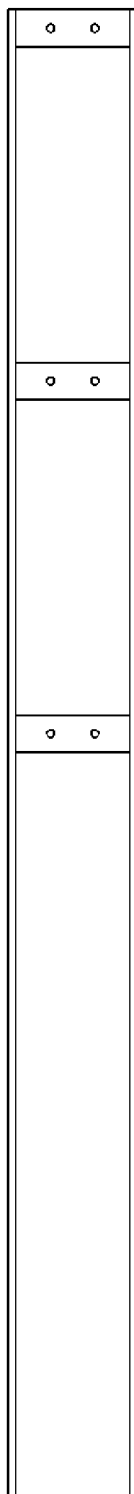


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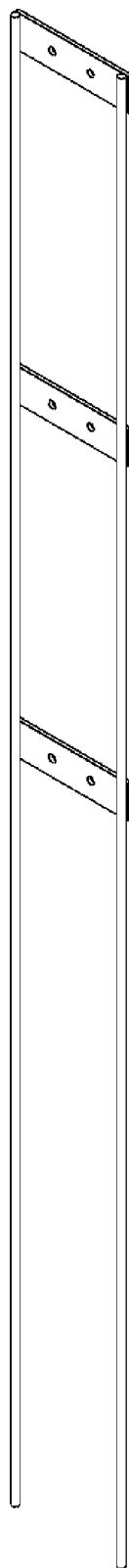


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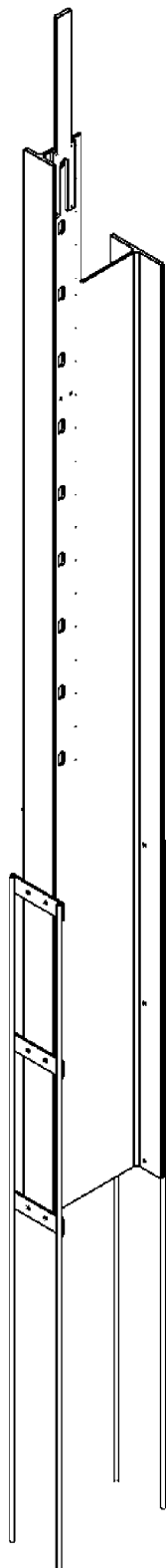


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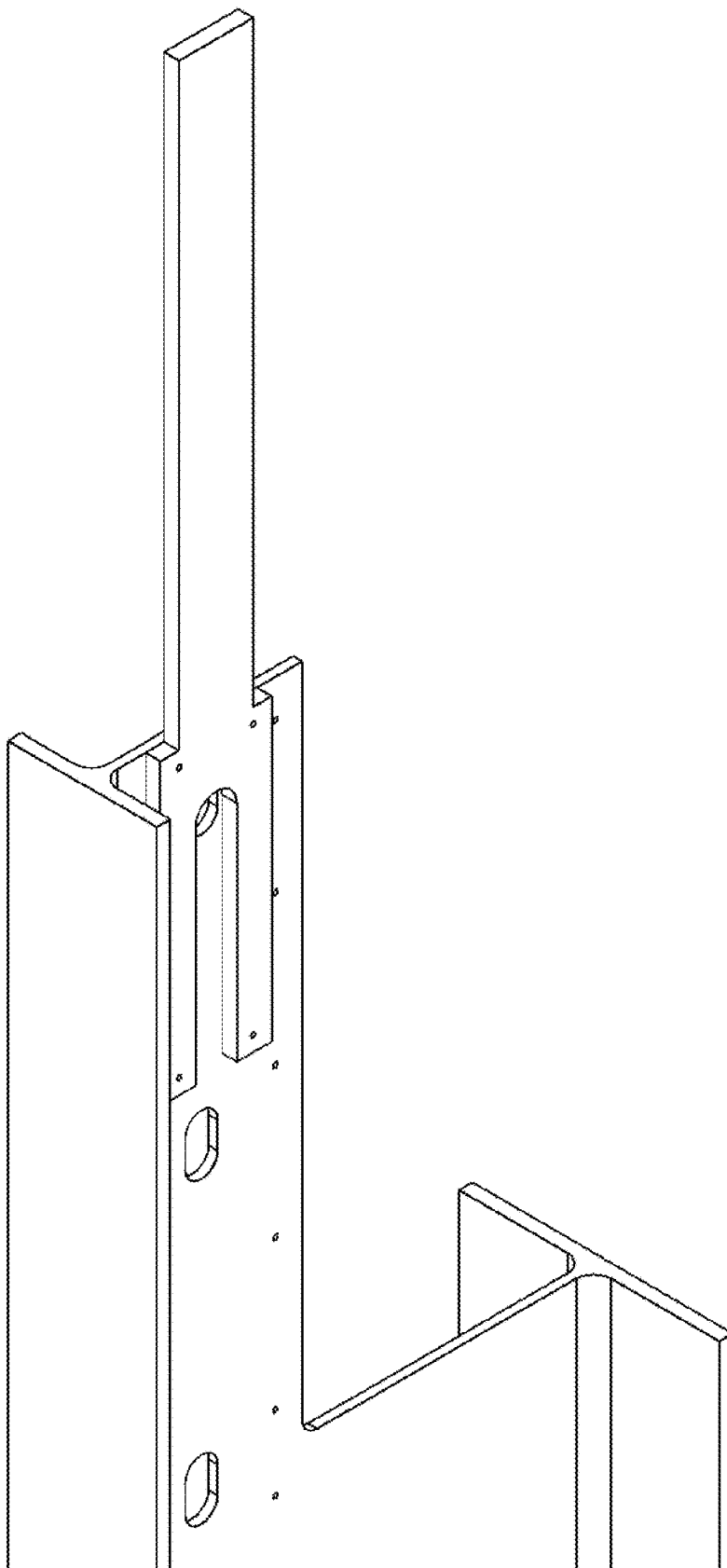


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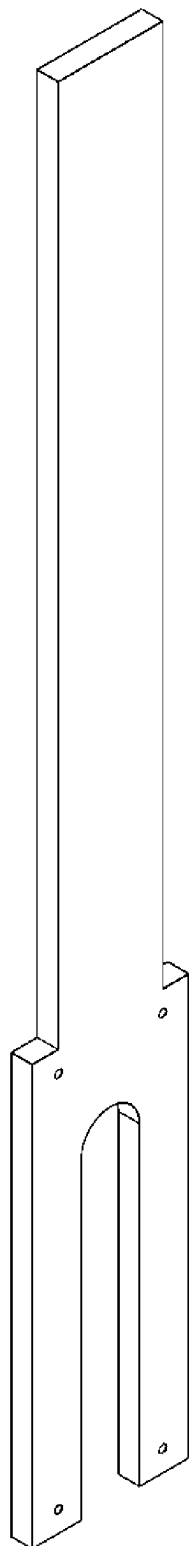


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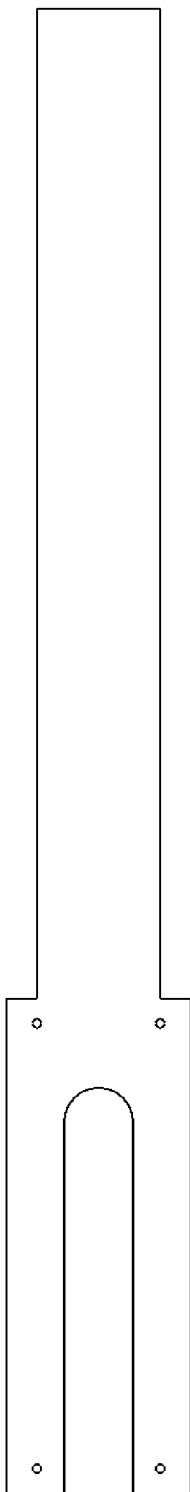


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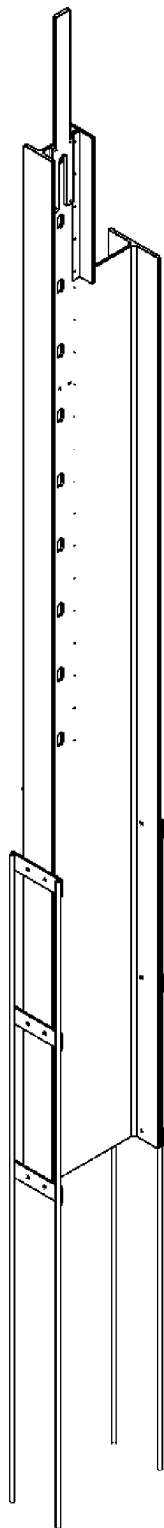


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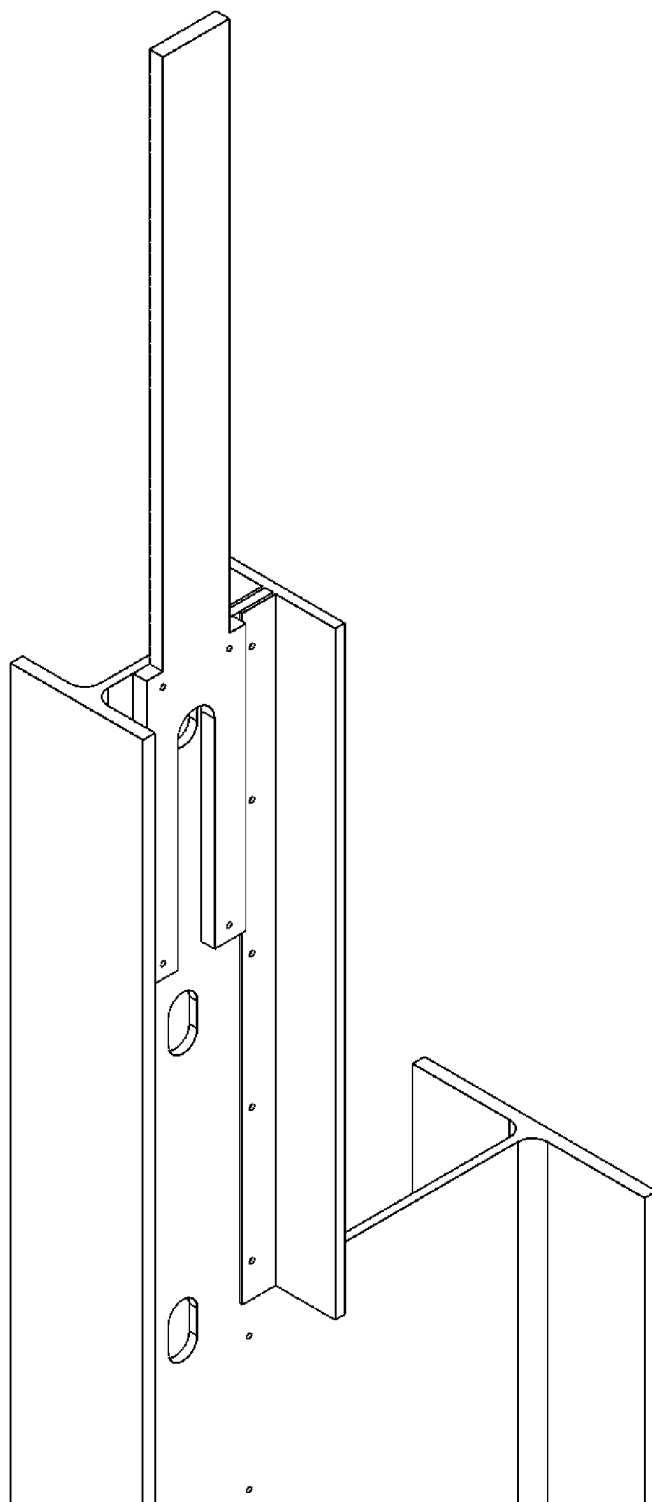


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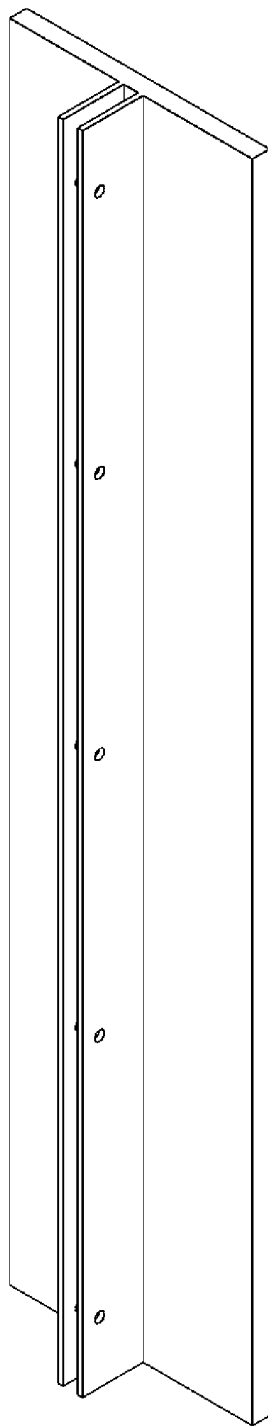


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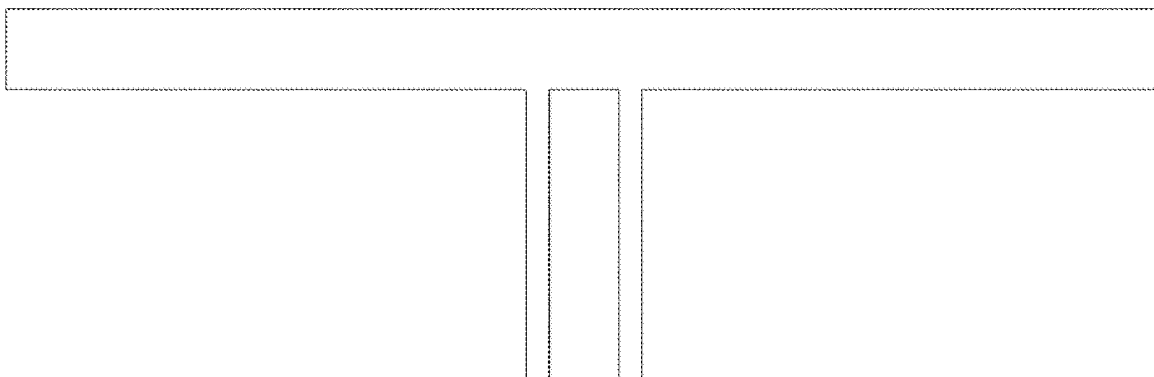


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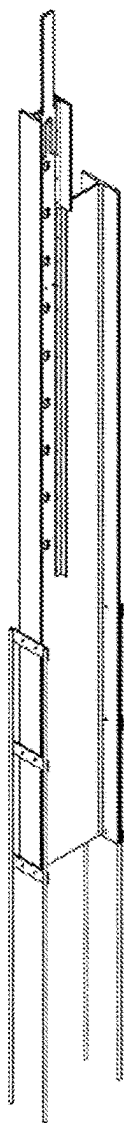


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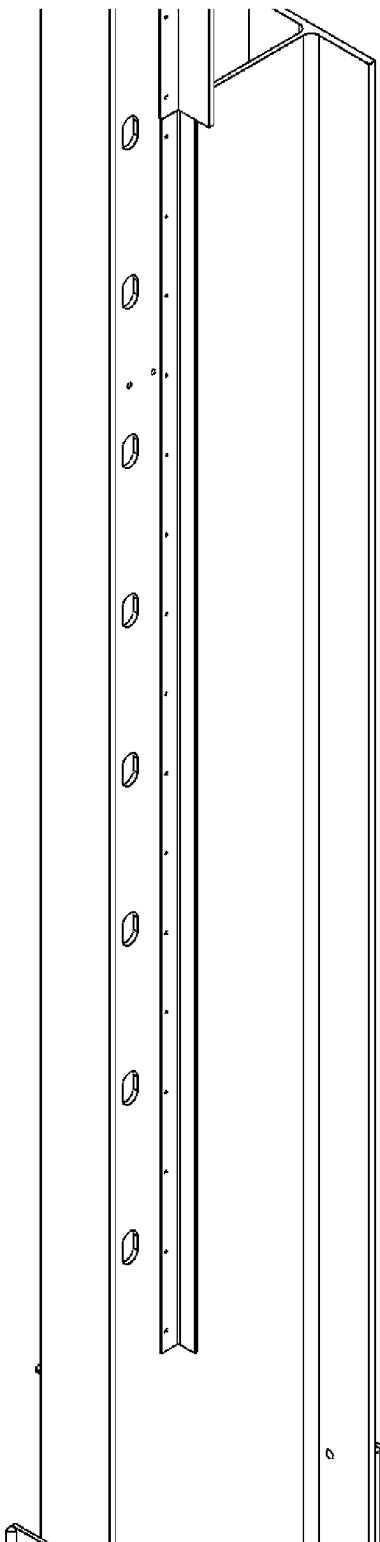


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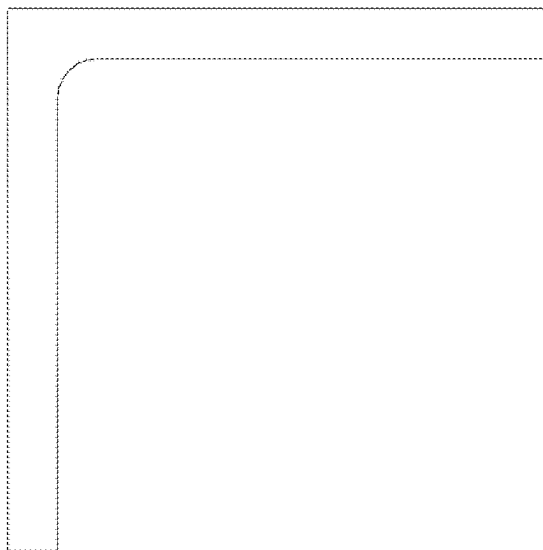


Figure 18.



Figure 19.

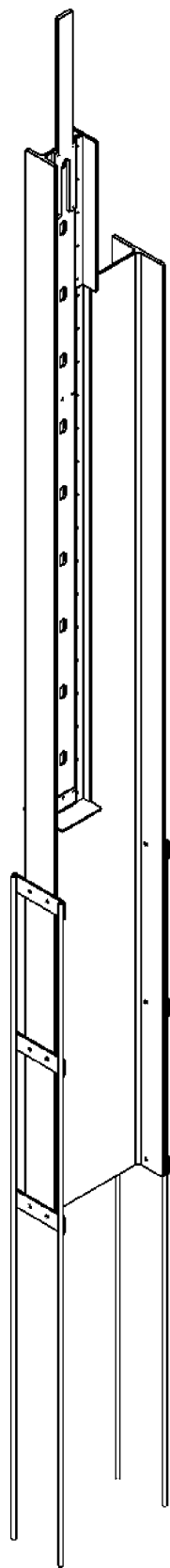


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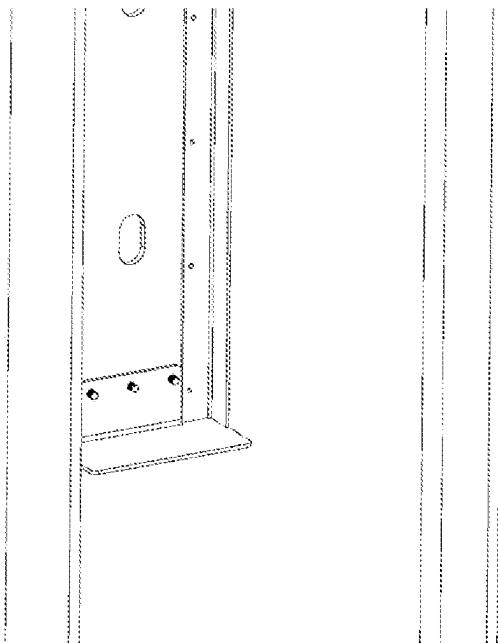


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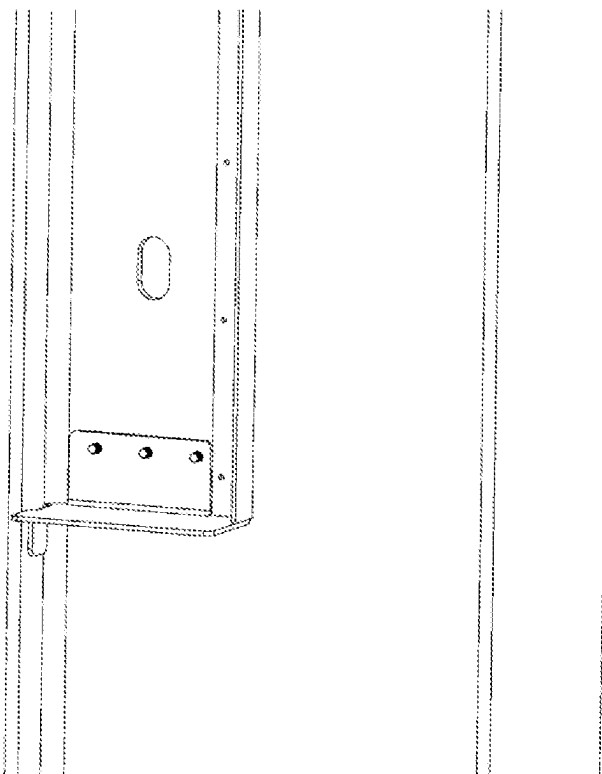


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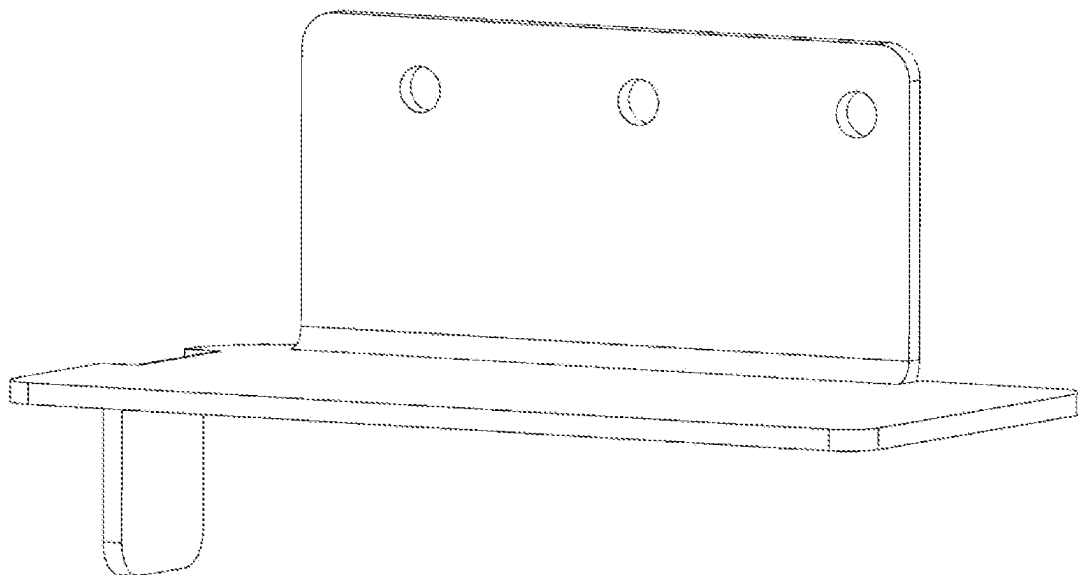


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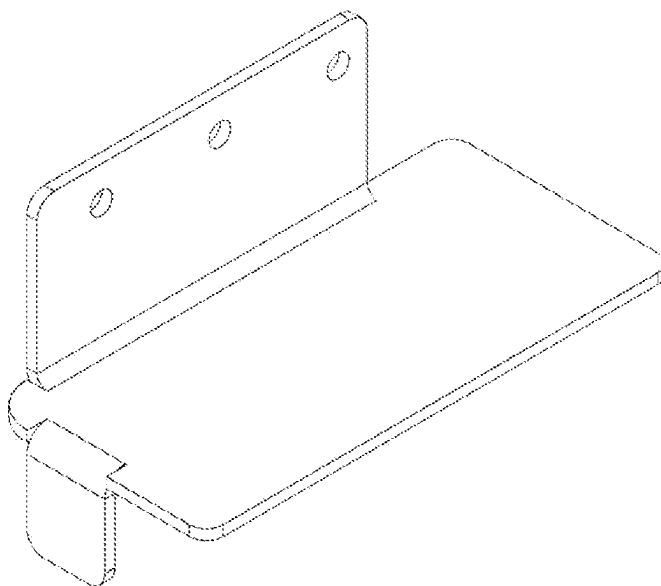


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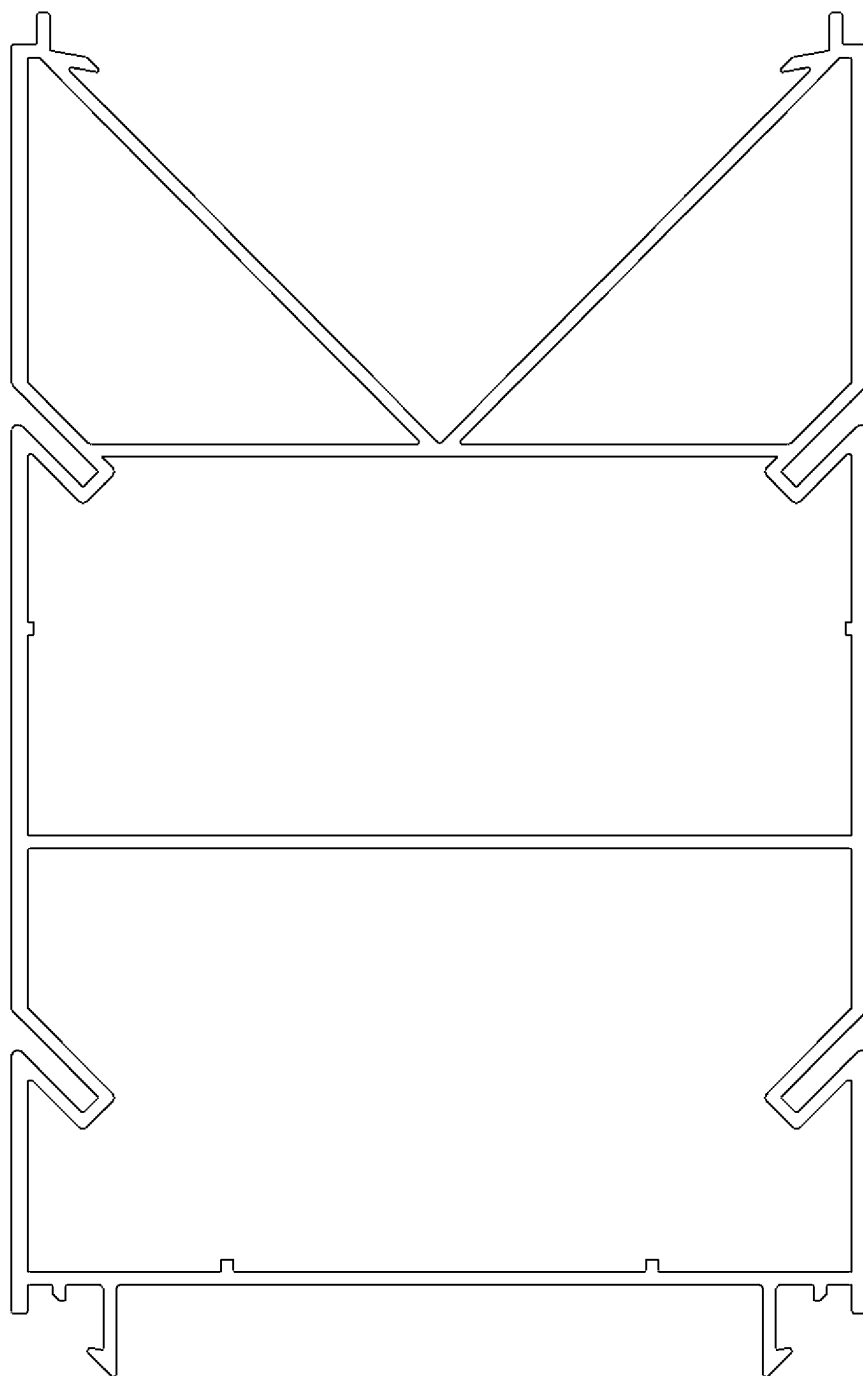


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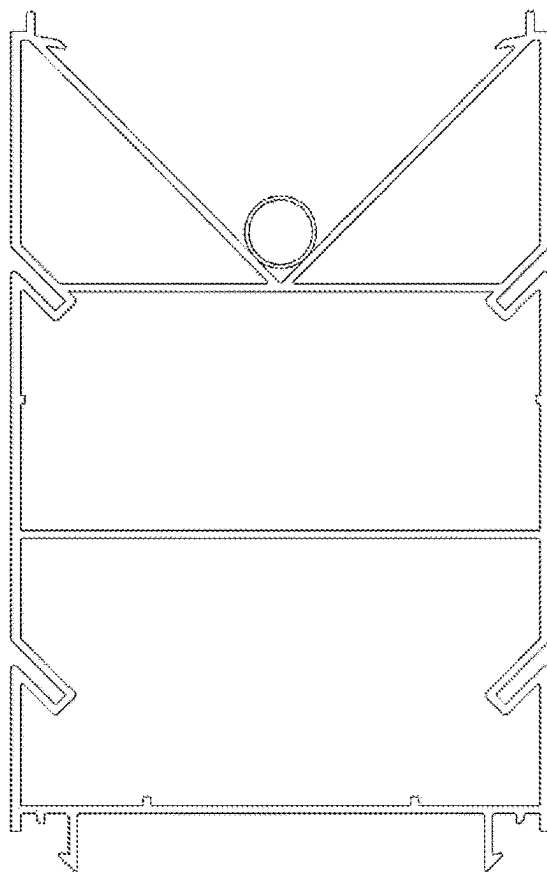


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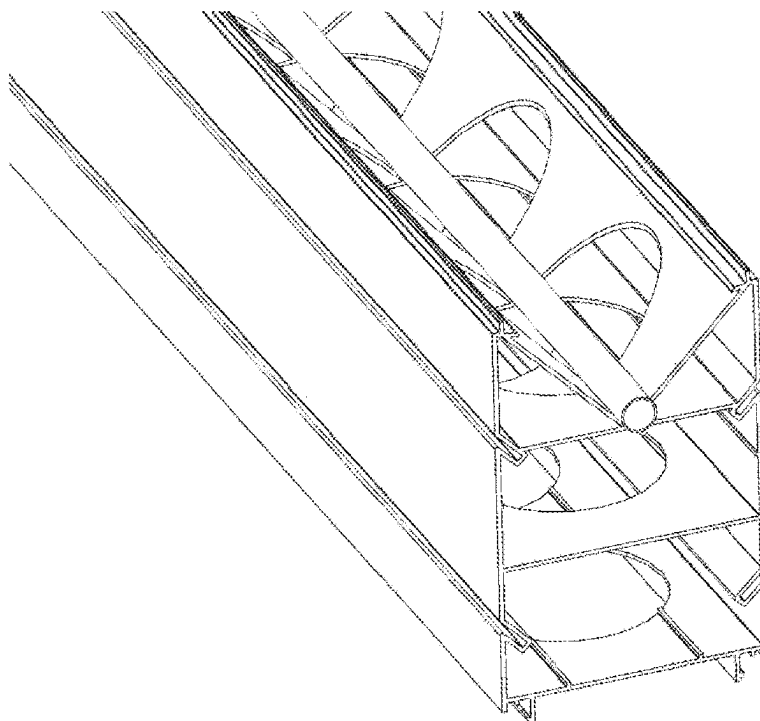


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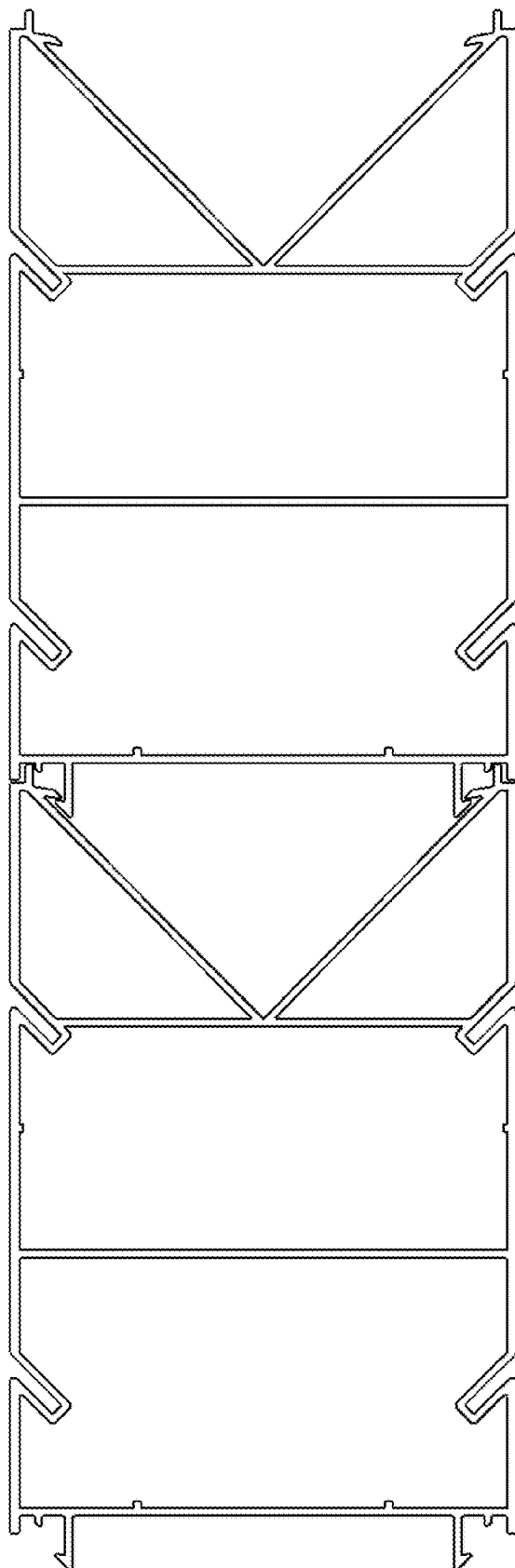


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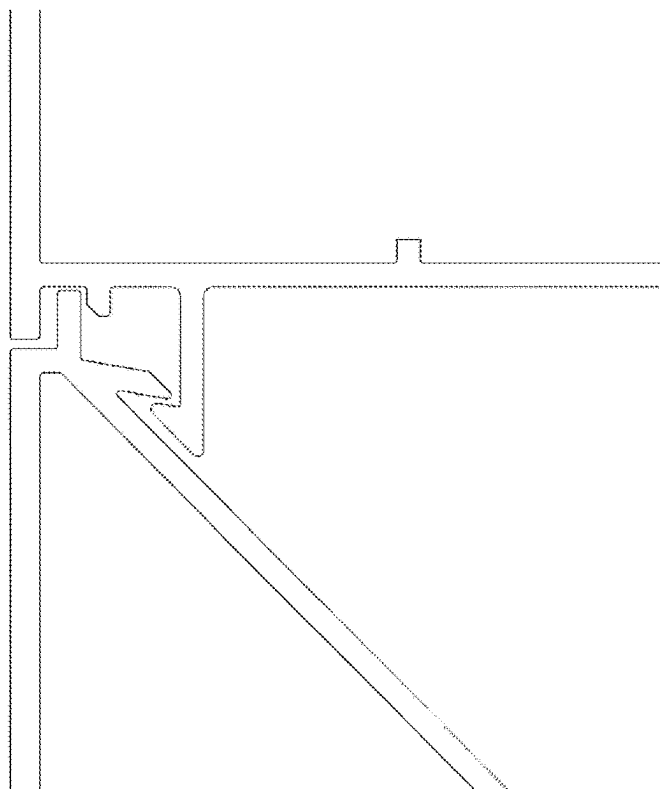


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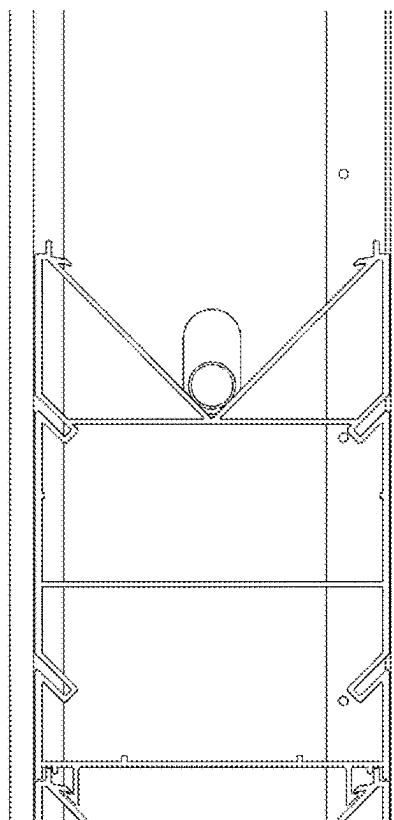


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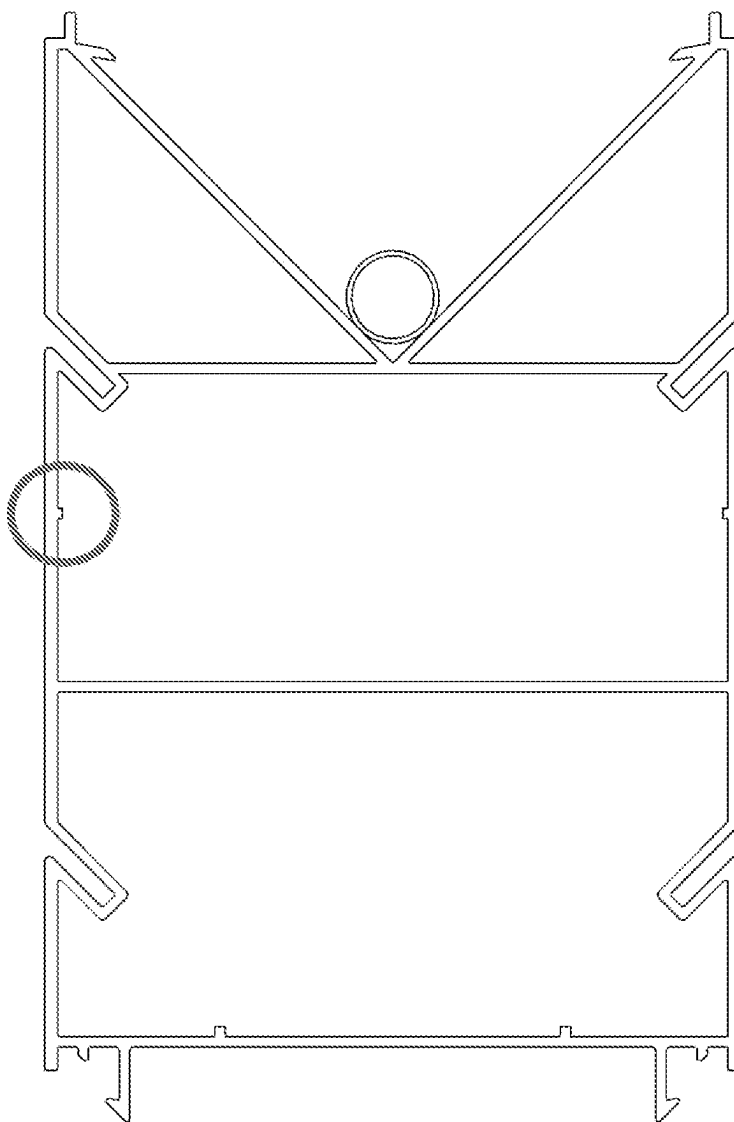


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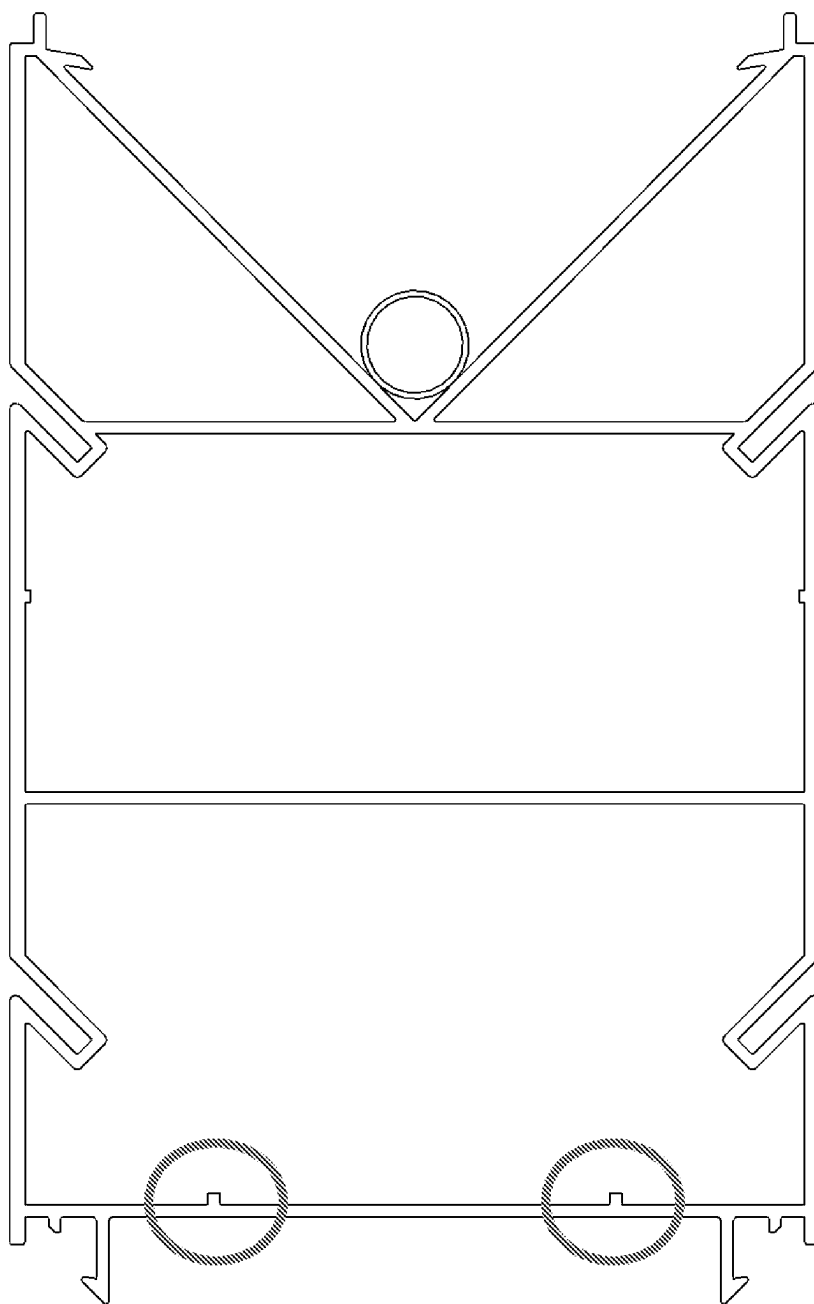


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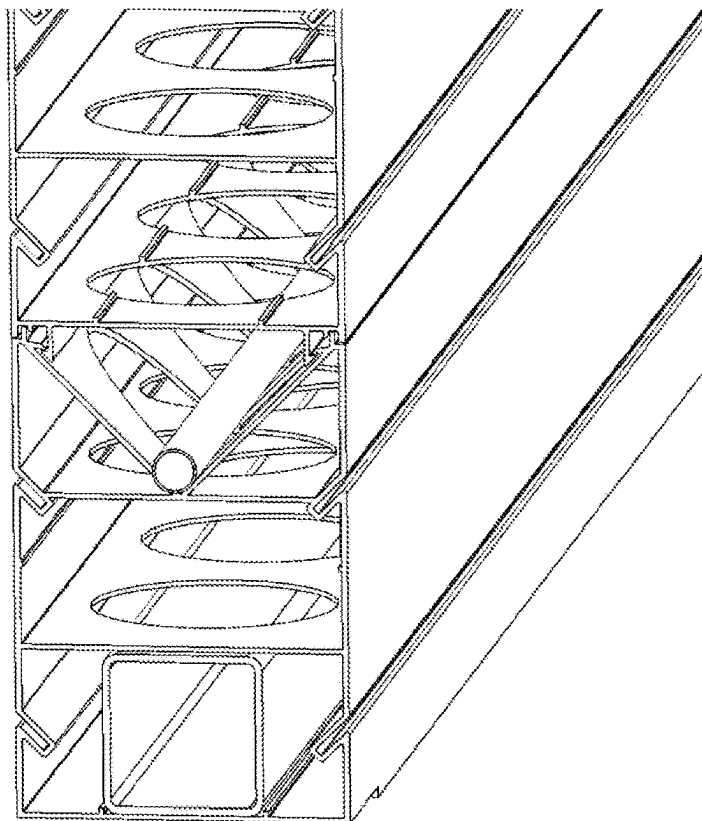


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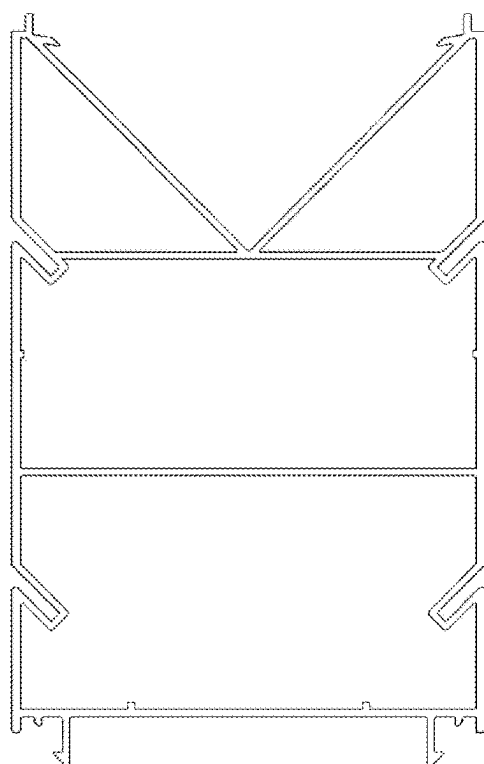


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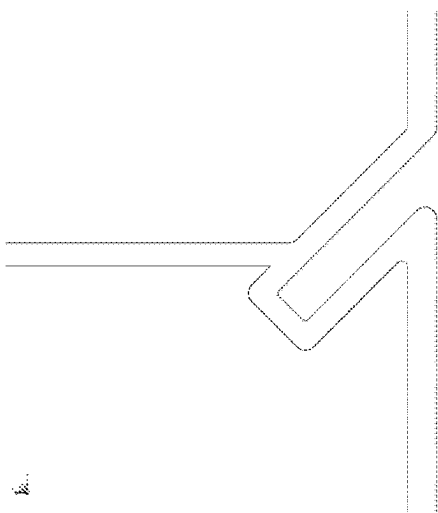


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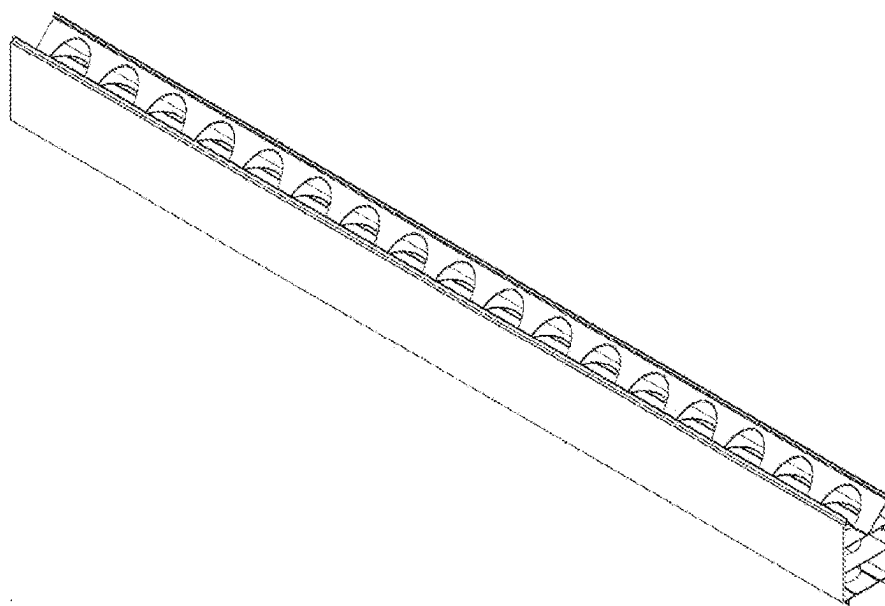


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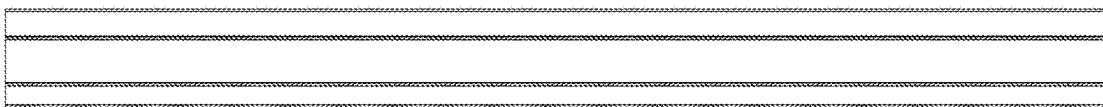


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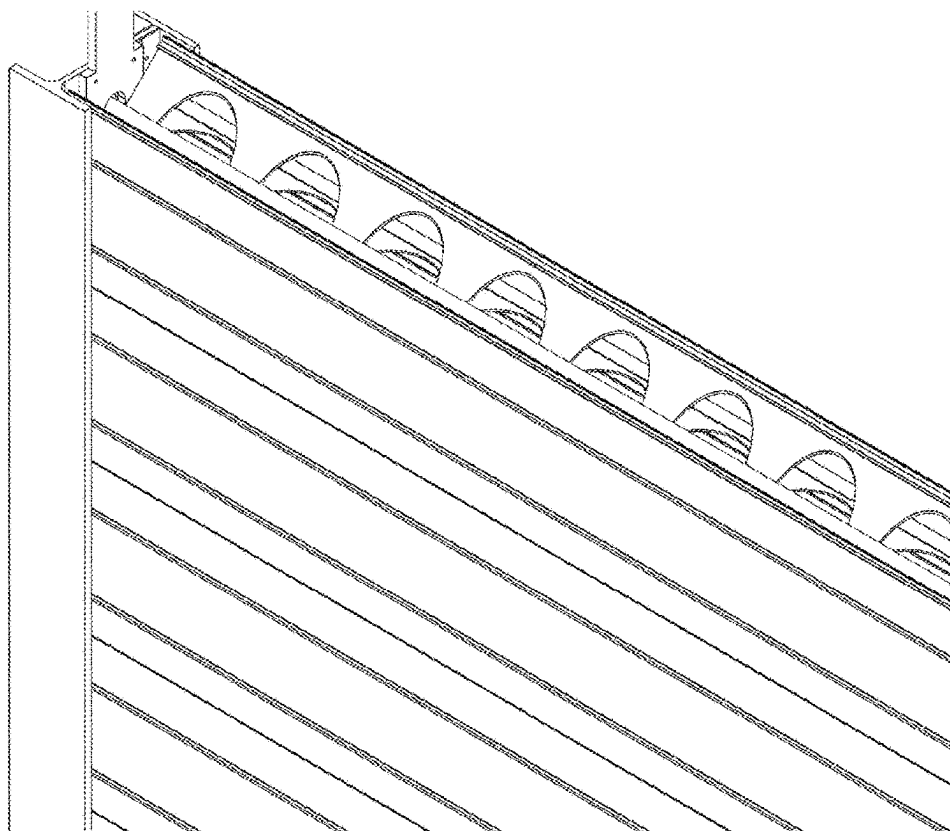


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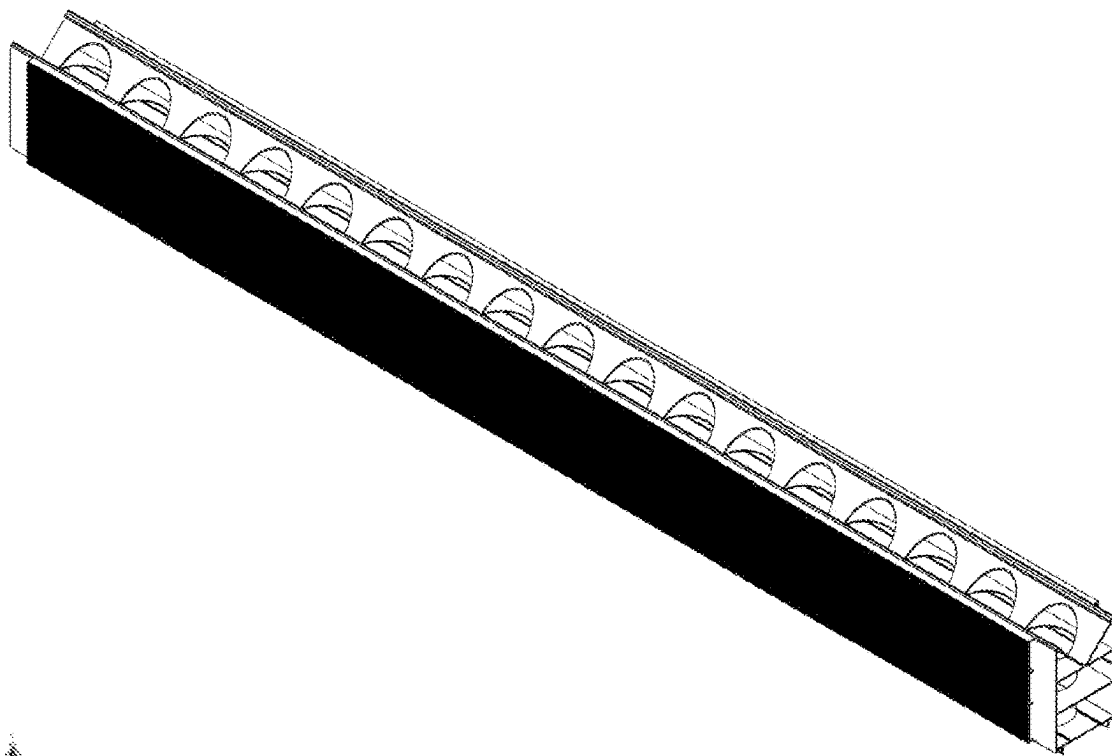


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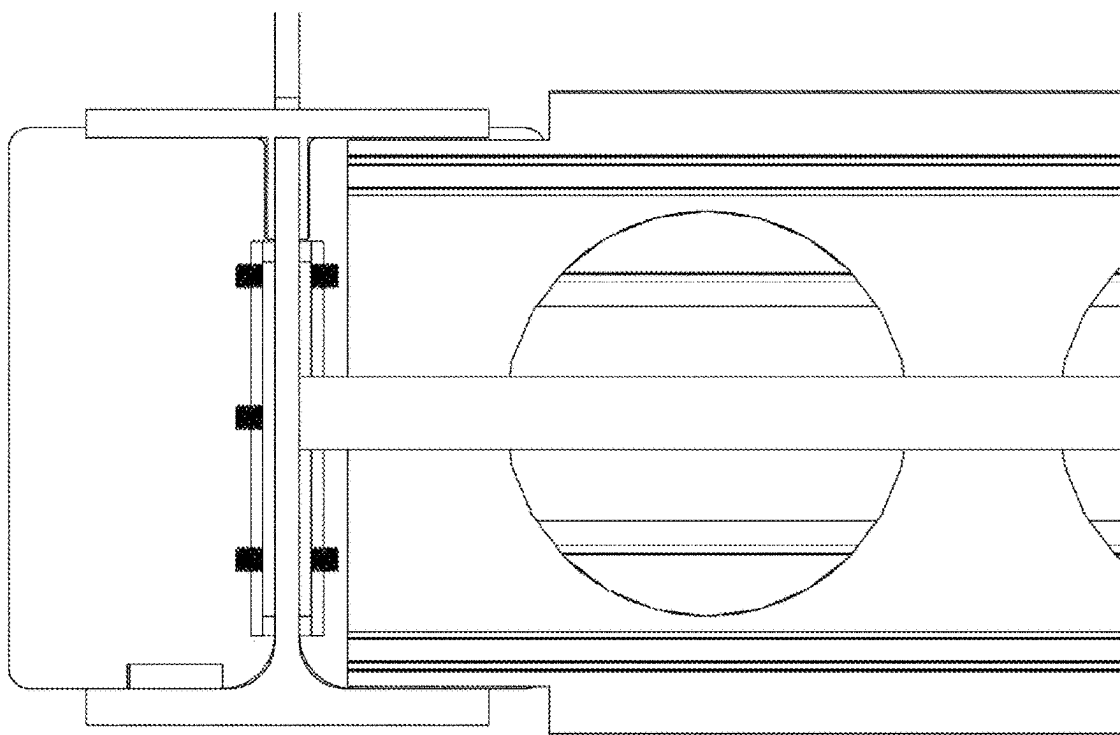


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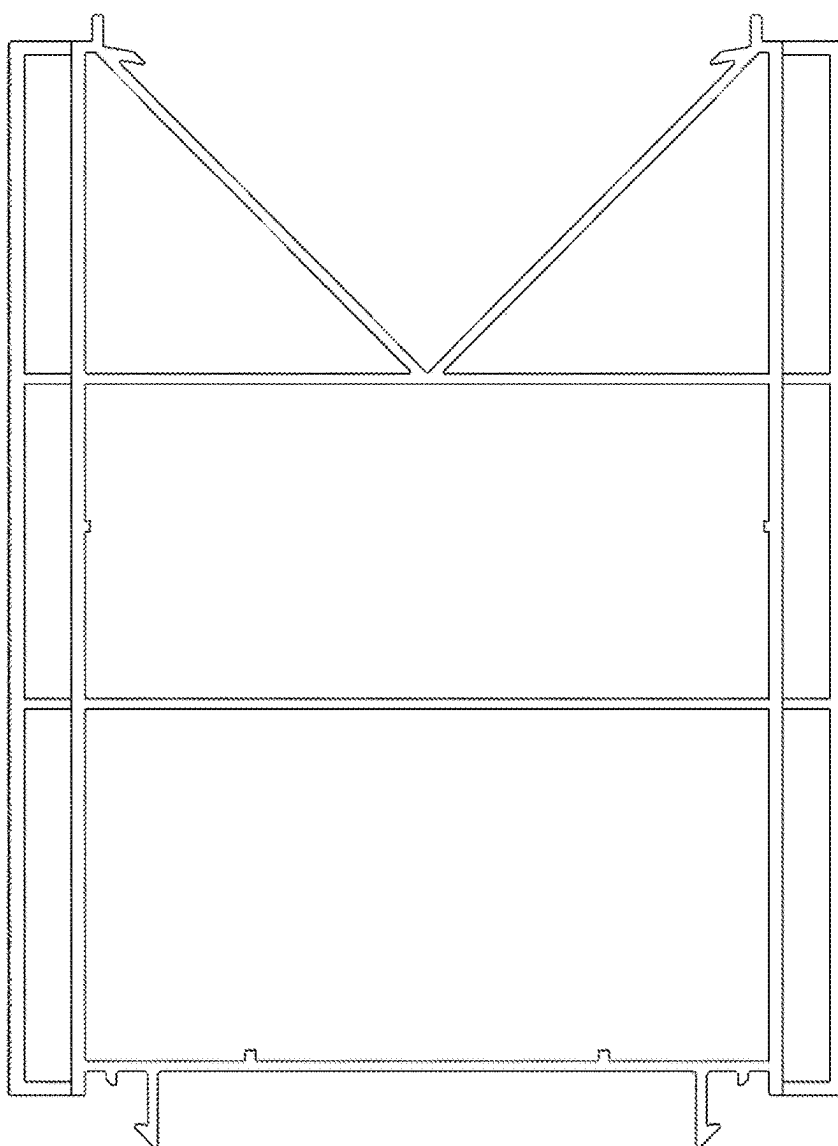


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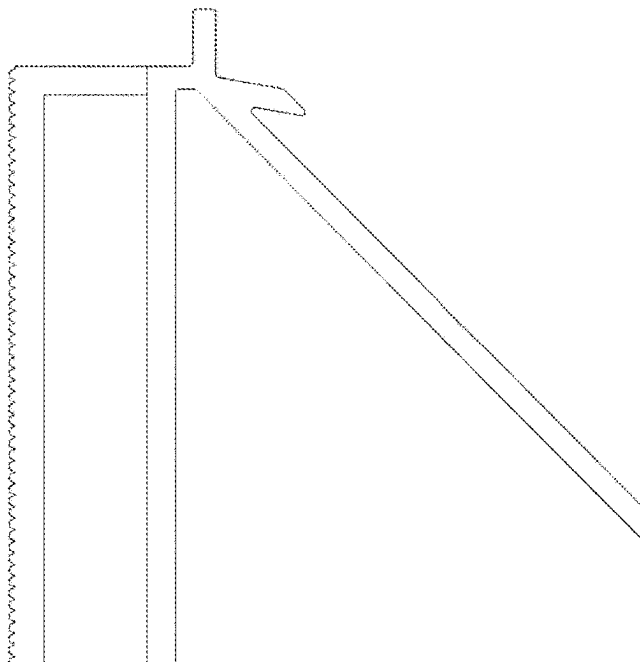


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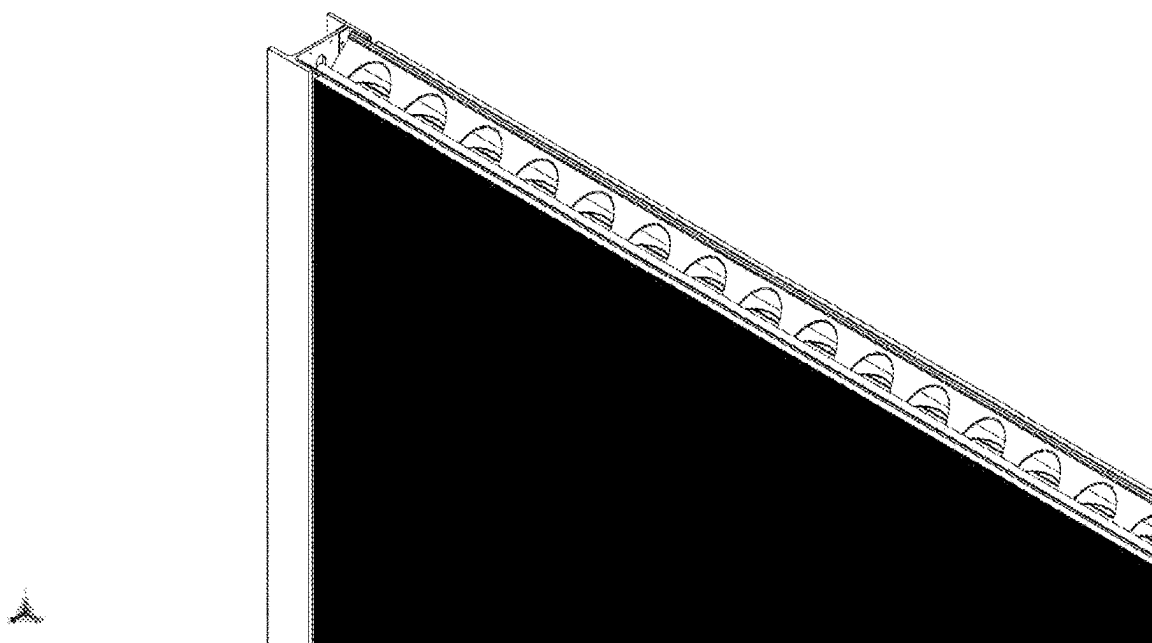


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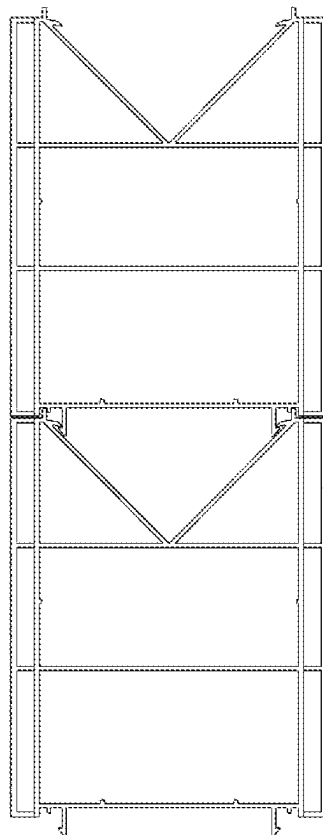


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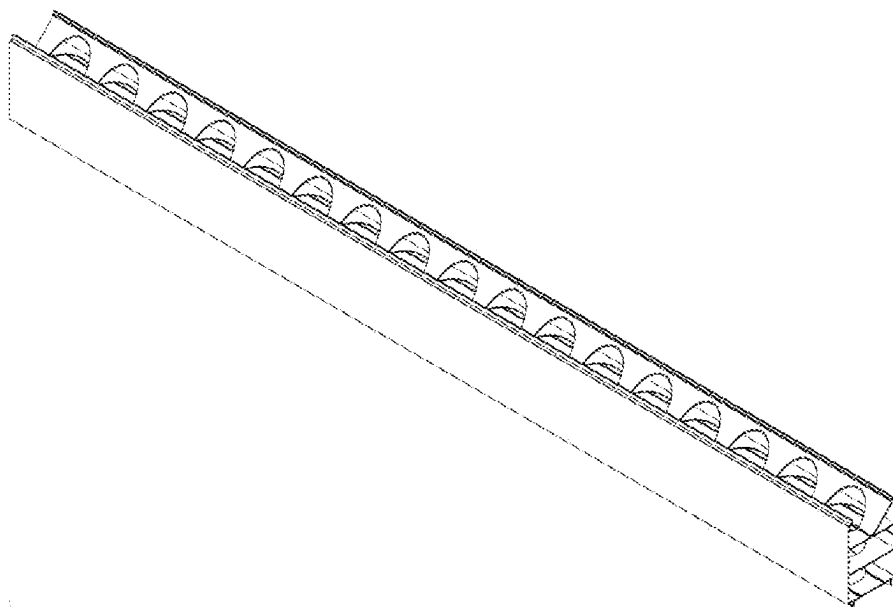


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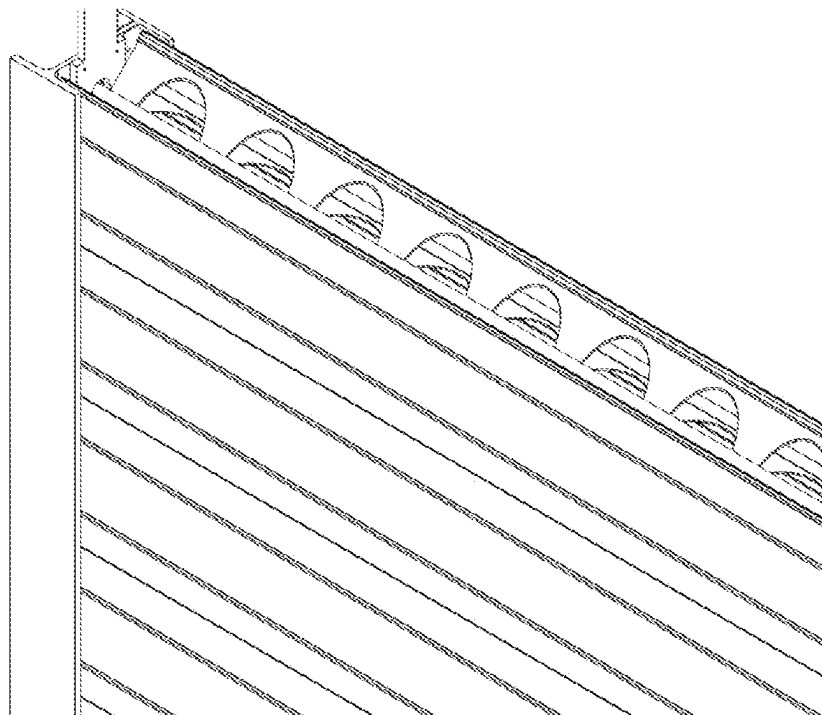


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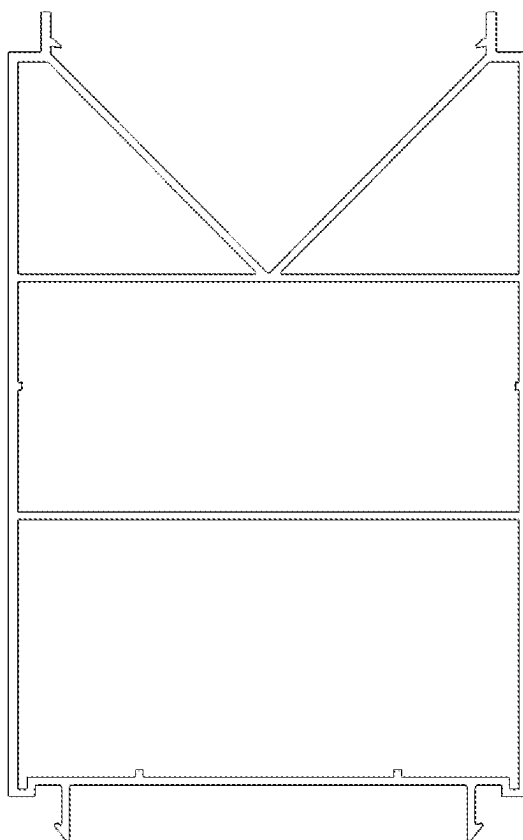


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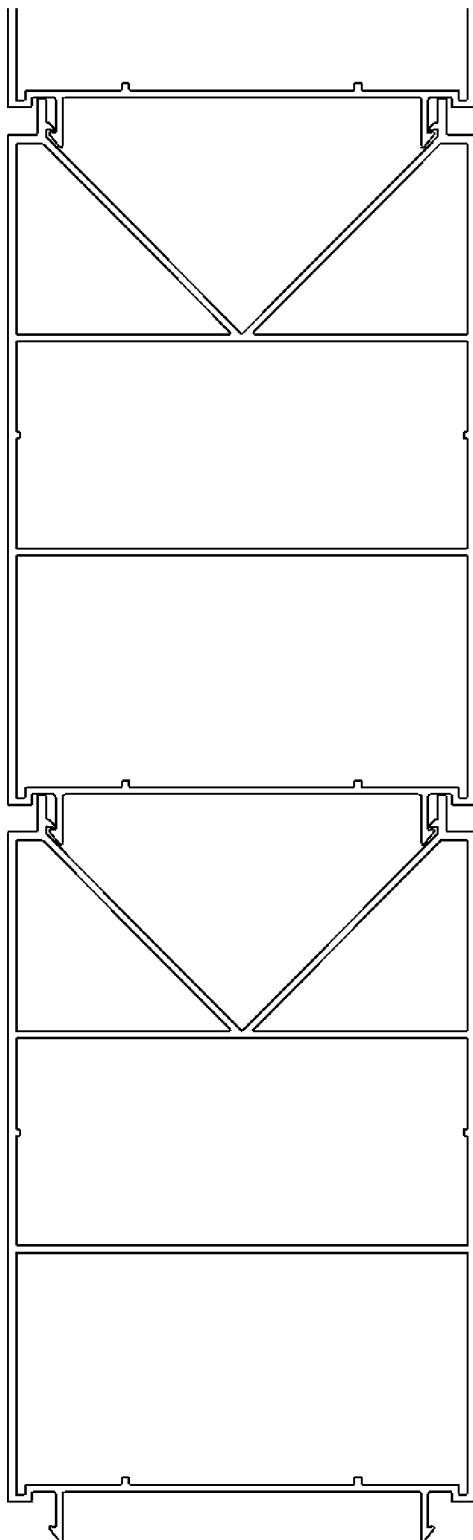


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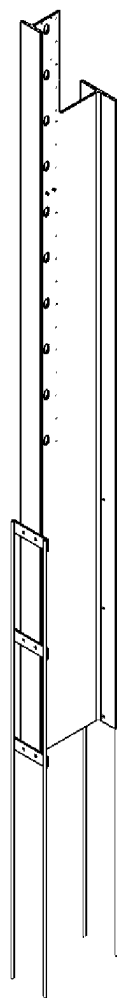
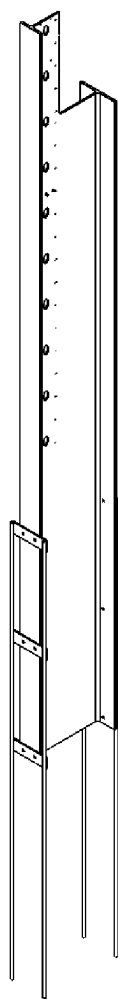


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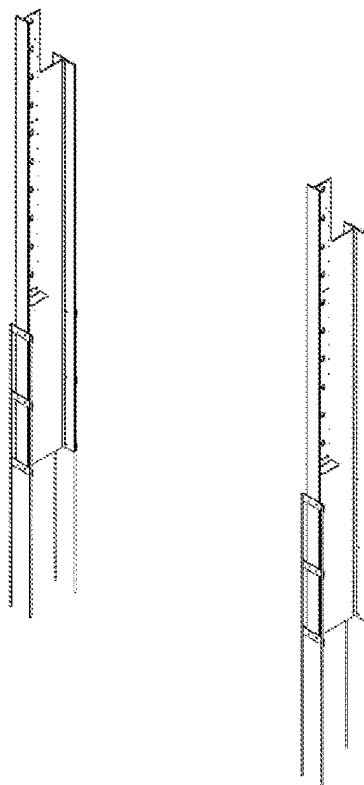


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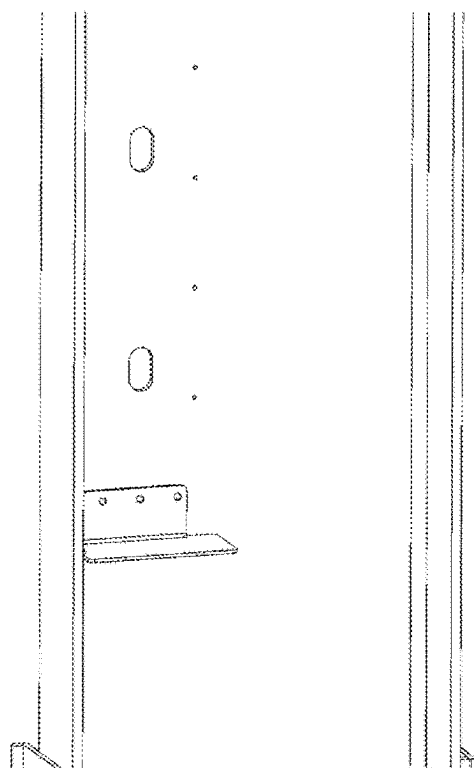


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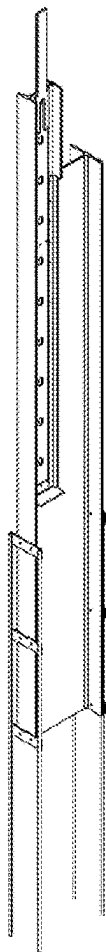


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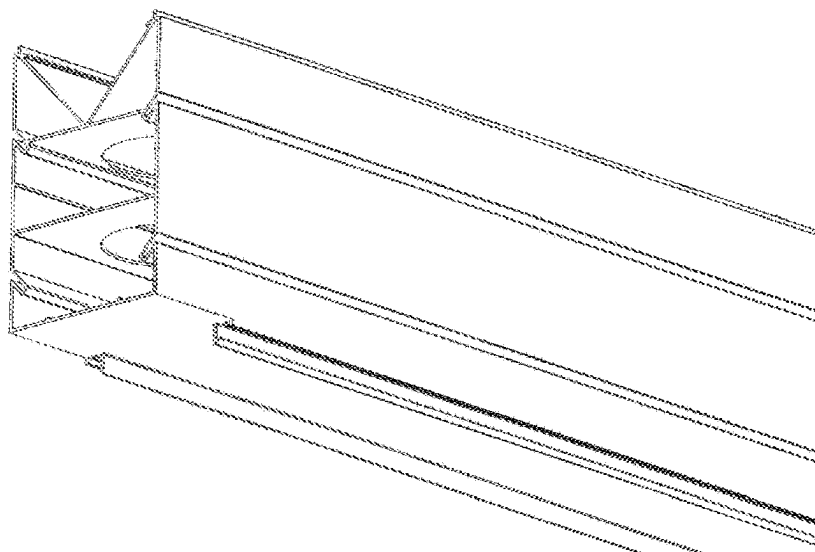


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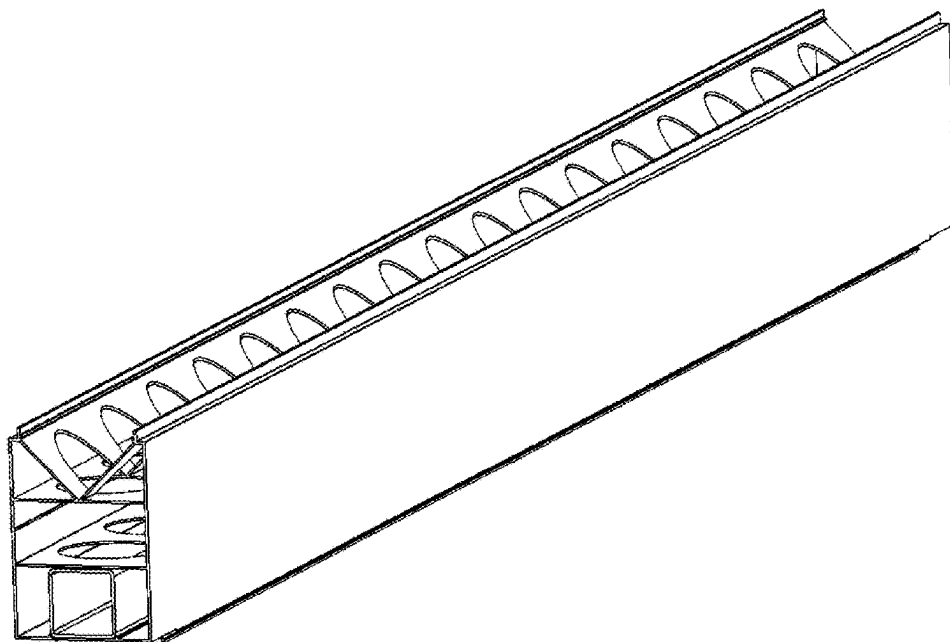


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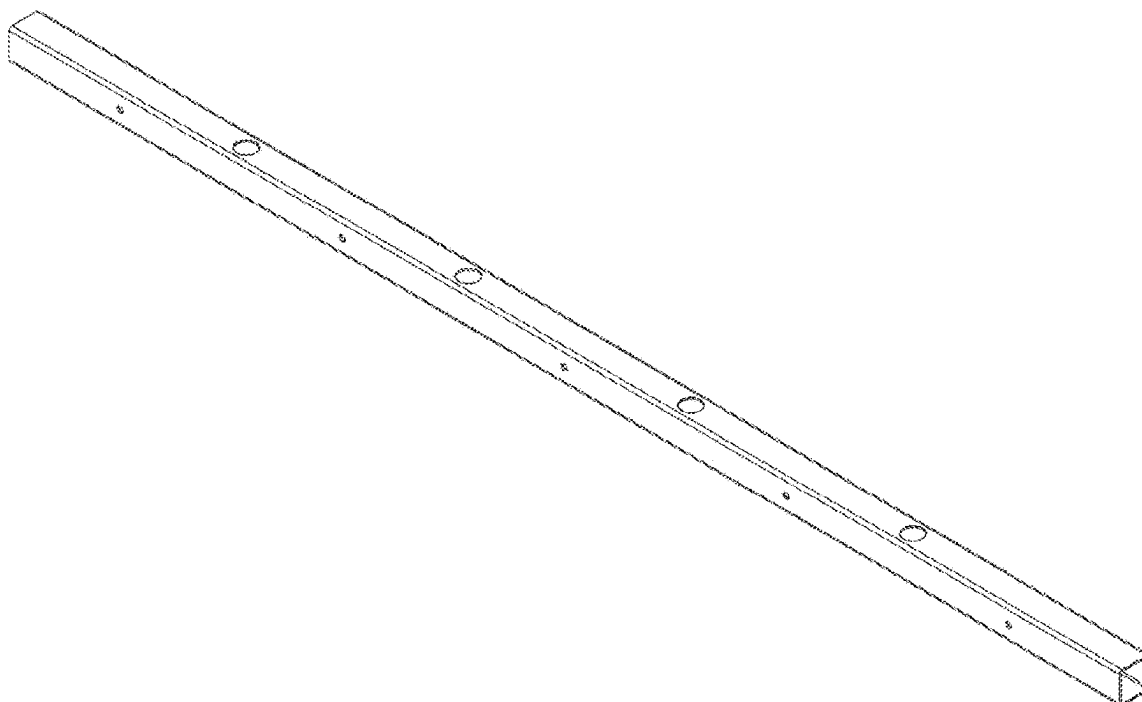


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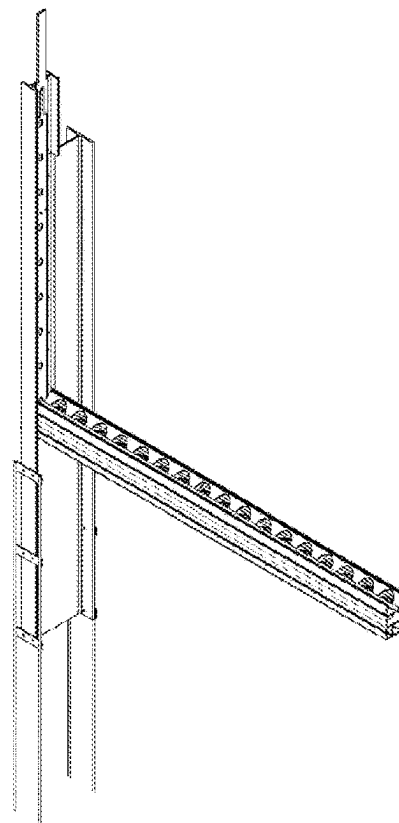


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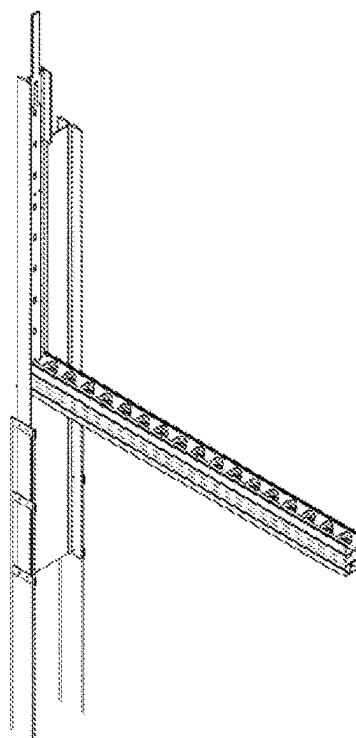


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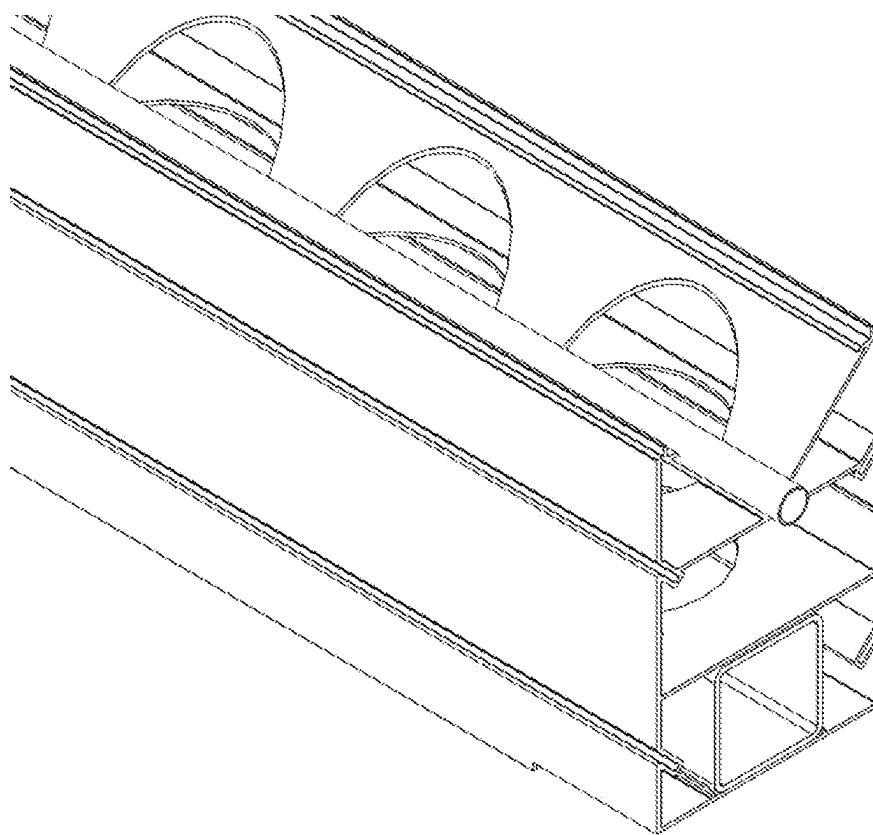


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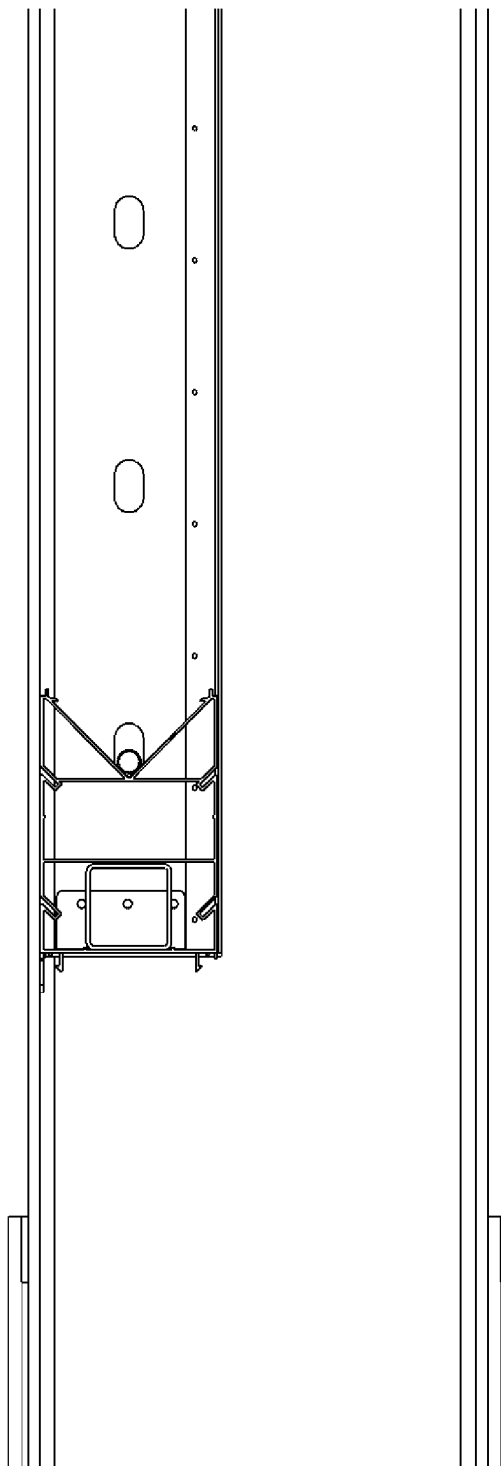


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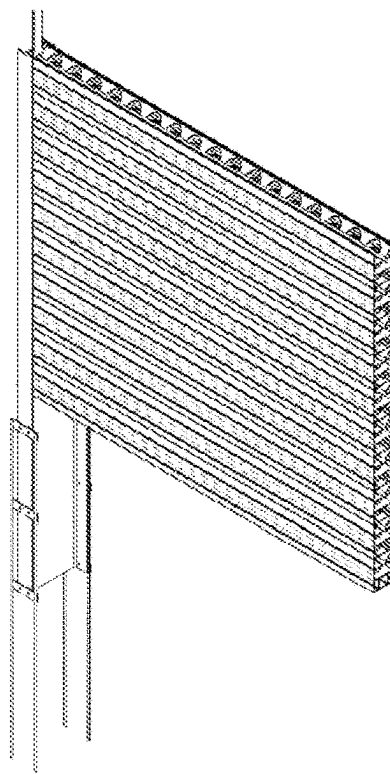


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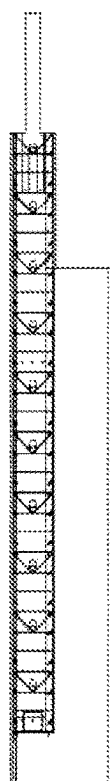


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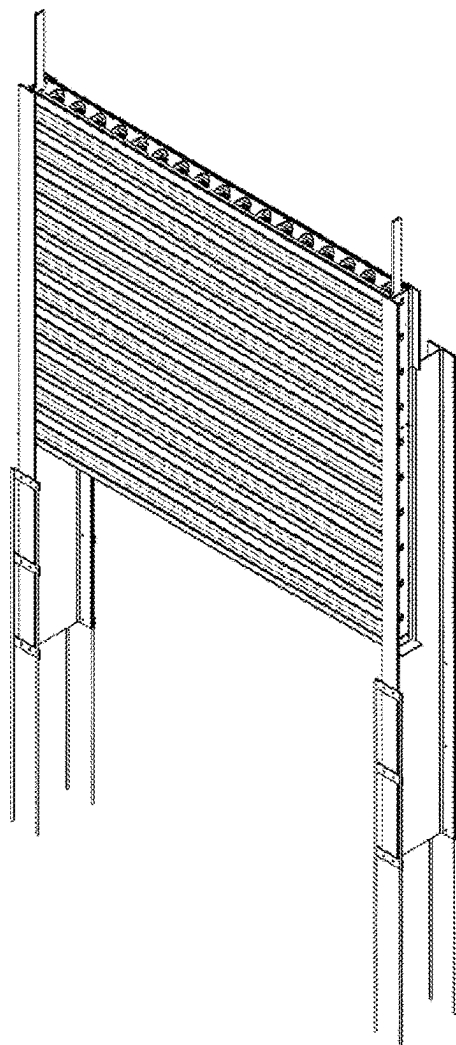


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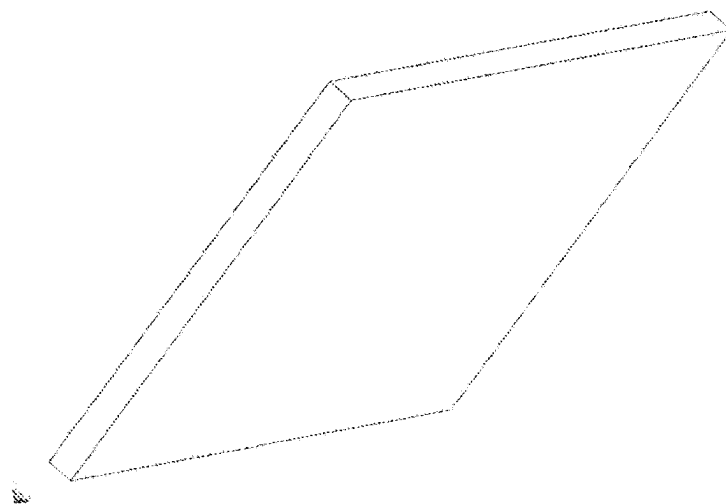


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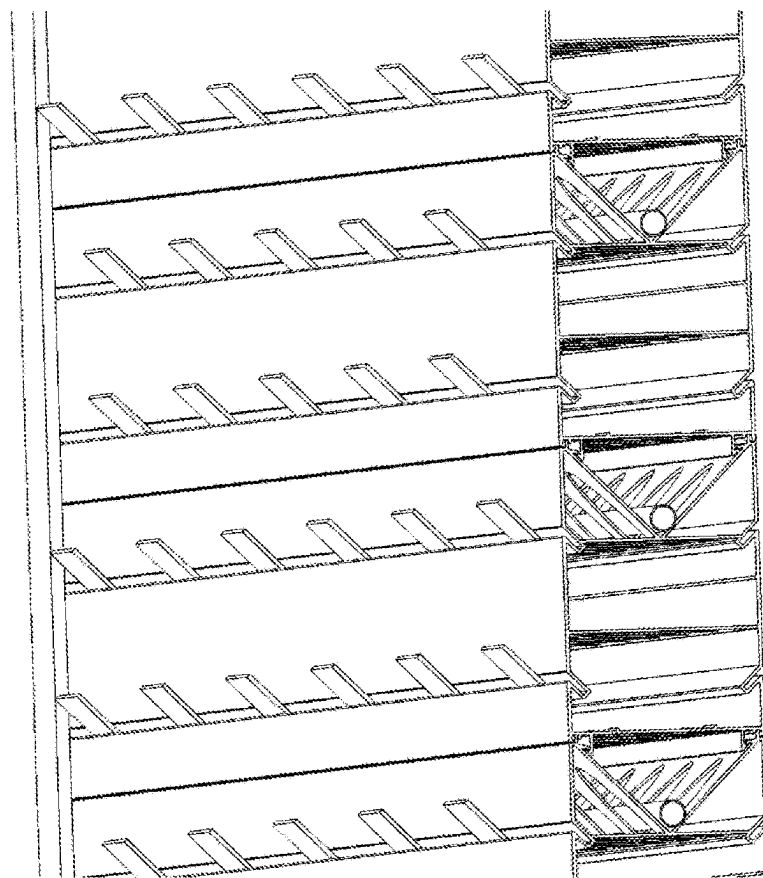


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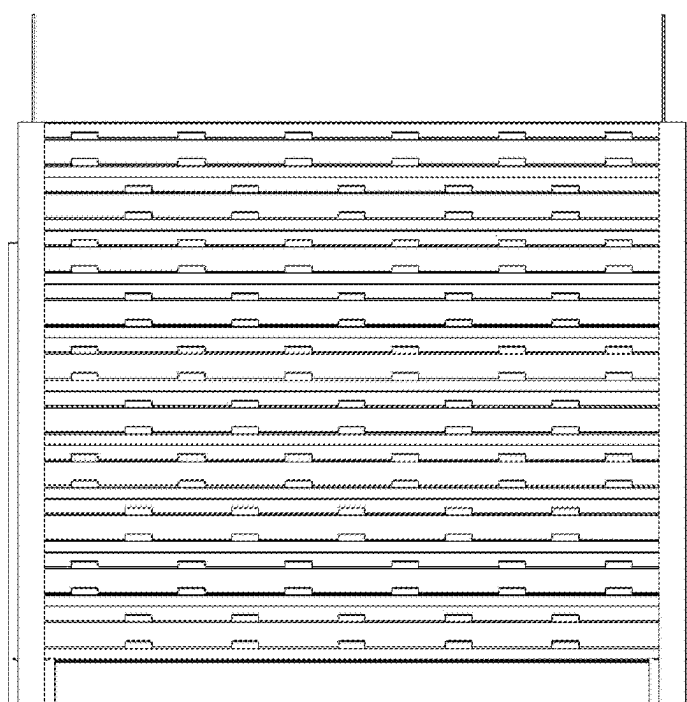


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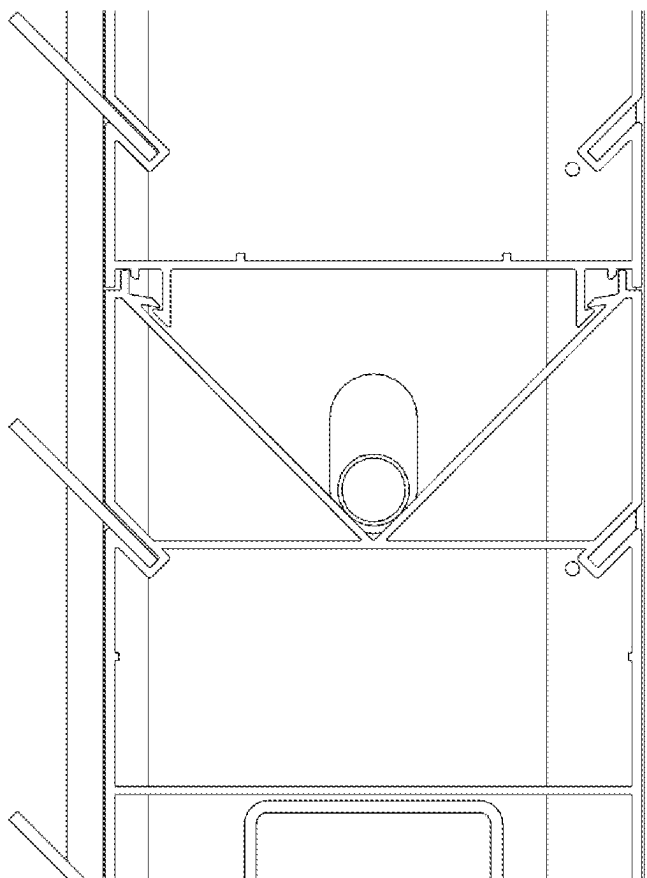


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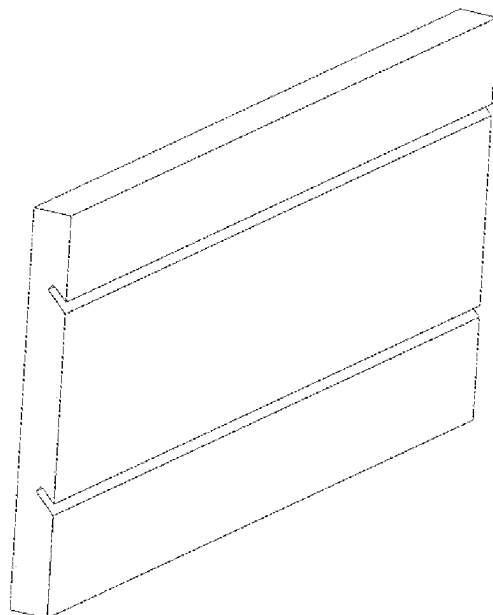


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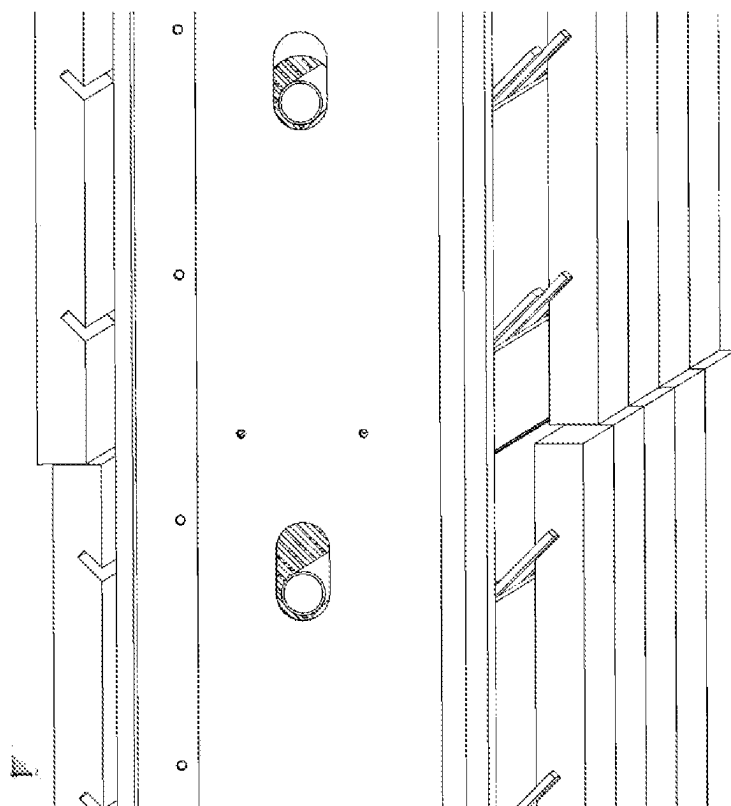


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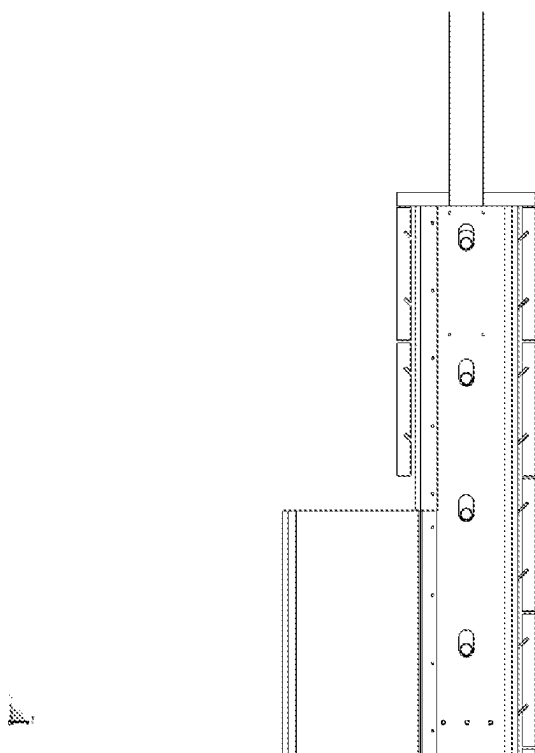


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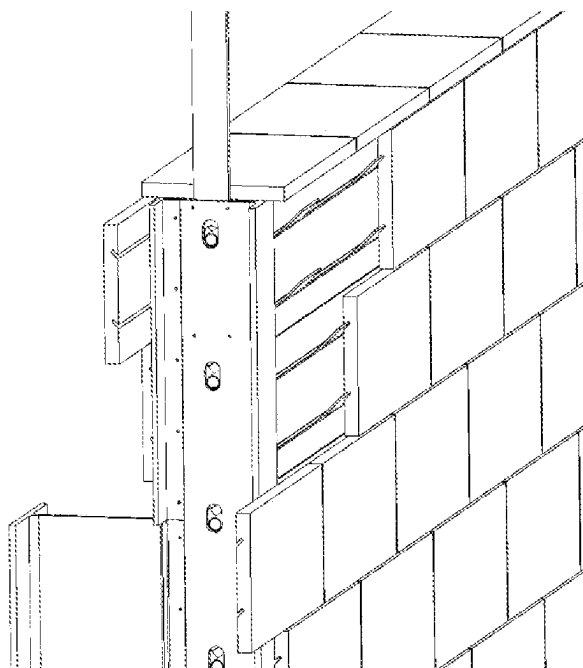


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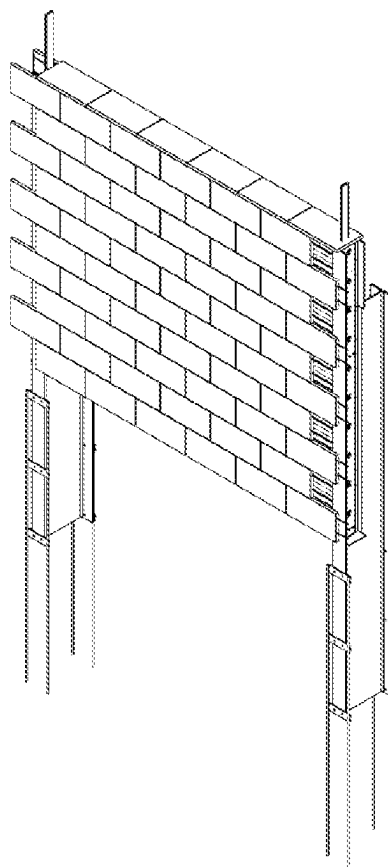
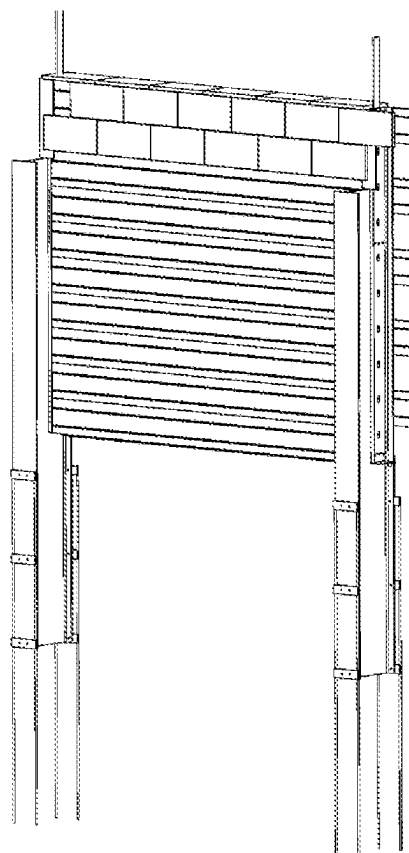


Figure 74.



FORMWORK FOR RETAINING WALLS

TECHNICAL FIELD

[0001] The invention relates to the field of construction systems for concrete walls, such as retaining wall for soil or earth and more particularly to those using formwork to form concrete, reinforced walls. The invention also relates to a modular formwork made from PVC (polyvinyl chloride) and other structural plastics.

BACKGROUND OF THE INVENTION

[0002] Concrete retaining walls are relatively rigid walls used for supporting soil laterally so that it can be retained at different levels on the two sides. Retaining walls are structures designed to restrain soil to a slope that it would not naturally keep to (typically a steep, near-vertical or vertical slope). They are used to bound soils between two different elevations often in areas of terrain possessing undesirable slopes or in areas where the landscape needs to be shaped for more specific purposes like hillside residential construction or roadway overpasses.

[0003] Retaining walls have been constructed via various known methods, including stone and concrete for many years. Concrete retaining walls require a formwork to act as a cast for the concrete. These have been made using many materials over time, including PVC.

[0004] The typical configuration for a modular polymer based retaining wall casting system is to align a succession of vertically open channels, defined by individually vertically aligned polymer casting elements, between vertical supports. The concrete is then poured into the vertical channels and allowed to set, thereby creating the wall.

[0005] In these prior art systems, if it is required to install reinforcing bars inside such walls, they must be inserted horizontally across the array of vertically aligned casting elements. They also require the installation and location of separate bar locators to assist in positioning the reinforcing bars.

[0006] Accordingly, it is an object of the invention to provide a modular retaining wall casting and system that improves or provides a useful alternative to the systems known in the prior art.

SUMMARY OF THE INVENTION

[0007] According to a first aspect of the invention, there is provided a waler adapted for use as part of a modular cast concrete wall system; said waler being adapted to interconnect with adjacent walers; said waler having side walls and upper walls defining an internal casting cavity; wherein said waler is adapted to be oriented in use in a substantially horizontal alignment by having relatively concrete impenetrable side walls and apertures in said upper and lower walls, thereby to permit passage of fluid concrete into and through said casting cavity.

[0008] Preferably, said waler incorporates an internal structure adapted to receive, and thereby operably locate, a reinforcing bar passing horizontally through said waler, and said internal structure is preferably integrally formed with the waler.

[0009] The waler according to the invention allows the wall casting structure to be assembled as a stack of horizontal components, each with a pre-formed holder for a reinforcing bar. This allows greater ease of construction and

flexibility of design. It also makes it easier to attach external features, such as cladding to the wall.

[0010] Preferably, said waler has an upper and lower section; said upper section defined by an open top wall narrowing to a V-profile adapted to receive said reinforcing bar; said lower section have in a profile approximating an enclosed rectangle.

[0011] It is preferred that said waler is constructed from extruded PVC, however many other suitable materials and methods of construction may be used.

[0012] In a preferred embodiment, there are provided rebates in the external wall of said waler that are adapted to allow attachment of external features, such as cladding to or the like, to the retaining wall.

[0013] In another aspect of the invention, there is provided a modular system for creating a concrete wall, incorporating a plurality of walers according to those described above. Preferably, said system incorporates a base waler that has a lower wall that is relatively concrete impenetrable.

[0014] Preferably, said base waler incorporates a reinforcing insert in said lower section, and said reinforcing insert is a square or rectangular metal tube having apertures in its upper and side walls to allow ingress of concrete.

[0015] Now will be described, by way of a specific, non-limiting example, a preferred embodiment of the invention with reference to the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIGS. 1 to 70 show various component parts, both assembled and individually, of the system according to the invention and the waler according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

[0017] The invention resides in the design and implementation of a modular concrete wall system that allows the modular casting elements to be laid horizontally, as opposed to the vertically aligned systems of the prior art. The individual walers according to the invention are designed to clip together top-to-bottom and the height of the wall is determined by the number of walers that are clipped together.

[0018] The example embodiment of the invention described below may be understood as a leave in place PVC formwork allowing for the in situ pouring of strengthened concrete walls. The walls can be free standing or used as retaining walls. The system according to the invention differs from the prior art in that its components are assembled by laying horizontally and connecting and stacking vertically rather, than the 'side-by-side' configuration typical of the prior art.

[0019] Load bearing is achieved by utilising soldier posts and piers rather than large footings and reinforcement. The unique profile of the walers according to the invention, and the system specific components, allow for relatively simple construction of the cast wall.

[0020] Turning to FIGS. 1 and 2, it can be seen that galvanised steel universal beams are utilised as posts for the system. Beams of specific dimensions and lengths are designated to each individual wall height. As required wall height increases, more heavily engineered posts are adopted. The posts are fabricated to allow for the connection of additional components and to allow for wall construction.

Oblong shaped holes provide for the through placement of reinforcing bar, as per FIG. 3.

[0021] Turning to FIGS. 4, 5 and 6, there are shown two lengths of deformed bar welded at their intersection across 3 pieces of steel flat bar.

[0022] For ease of installation and transportation all universal beam post lengths preferably measure under 6 metres in length. Where post lengths greater than 6m are required to obtain sufficient embedment in ground, a post extension is added at opposing sides to the base of posts. The post extension is preferably fixed using a bolted connection.

[0023] Turning to FIGS. 7, 8, 9 and 10, there are shown fence brackets, typically cut from plate steel (FIGS. 9 and 10). Should fencing be required atop of the wall, fencing brackets are added as per FIGS. 7 and 8. Brackets can be placed on either side of posts and will be encapsulated by concrete when core filled. Voids in each bracket allow for reinforcing bar through-placement.

[0024] Turning to FIGS. 11, 12 13 and 14 there is shown a post backing plate, typically made from extruded PVC. The post packing plate (FIGS. 13 and 14) is attached to the cut away area of the beam post (per FIGS. 11 and 12) to maintain waler placement and allow concrete retention when core filling.

[0025] Turning to FIGS. 15, 16, 17 and 18, there is shown a backing angle, typically made from extruded PVC. The backing angle (FIGS. 17 and 18) is fixed, as per FIGS. 15 and 16, to maintain waler placement and to allow concrete retention when core filled.

[0026] Turning to FIGS. 19, 20, 21, 22 and 23, there is shown a pos base plate, typically made from folded plate steel

[0027] The flat surface of the base plate is fixed at the bottom of wall (BOW) level, supporting the bottom waler in the correct location and allowing concrete retention when the waler is filled. The base plate includes a folded tab so as able to be clamped to the front of the post when positioning. A single base plate is located and clamped to one side of the post and holes are drilled through the post corresponding to the plate hole locations. A second plate is positioned over the generated holes on the alternate side of the post and then both plates are fixed using a bolted connection. This process allows for a constant height level to be carried through for walers entering and leaving the post.

[0028] Turning to FIGS. 24, 25, 26, 27, 28, 29, 30, 31 and 32, where is shown the waler according to one aspect of the invention.

[0029] The horizontal interlocking PVC extruded waler sections form the main part of the wall. In this example, the walers are extruded PVC pieces with intermittent holes bored vertically through the piece along its complete length for the ingress of concrete. The walers span between the universal beam posts described above and are adapted to be core filled with a high slump, high strength ready mixed concrete.

[0030] The features common to all walers within the system are a V shaped open top and an inward functioning click together joint located at the top and bottom of the waler, allowing simple attachment to duplicate walers above and below. Other features include fill level line and SHS positioning protrusions.

[0031] The V-shaped open top of waler according to the invention provides structural integrity to the component while allowing access to manoeuvre structural steel rein-

forcing bar (shown in position in FIGS. 25 and 26). The reinforcing bar is naturally located at the base of the V joint in appropriate position. Prior art PVC formwork examples are closed on the perpendicular at each end and require additional mechanisms for bar positioning.

[0032] The 'click together' joint mechanism, provided by complementary protrusions on the upper and lower surfaces of the waler shown in detail in FIG. 28, allows for stacked pieces to be secured to the piece underneath and above, as per FIG. 27.

[0033] Walers are sleeved at each end by the post which inhibit expansion outward. The joining mechanism allows for an inward functioning connection. The joint allows for connection using downward pressure on directly vertical added pieces. Most prior art PVC formwork uses outward expanding joints.

[0034] A fill level line (marked in FIG. 30) is a raised nodule of PVC that is positioned 100 mm from the base of the waler. This indicates the optimum level of grout fill during the initial core filling process.

[0035] Positioning protrusions (marked in FIG. 31) allow for the square hollow section (SHS) reinforcing bar insert to be centrally located in a base plate, as per FIG. 32. The SHS includes holes along its top and side axes to allow the free flow of grout when core filing the waler.

[0036] Turning to FIGS. 33 to 50 it is noted that three forms of waler will are typically offered with the system. While similar in their form and functionality, they differ slightly depending on the wall finish required.

[0037] When the completed wall is to be clad, a clad waler (FIG. 33) is used. The waler includes horizontal grooves (shown in detail in FIG. 34) running continuously along its full length, per FIG. 35. The grooves allow for the quick location and positioning of a plate fixing for the cladding application.

[0038] When the wall is to be rendered, a rendered waler (FIGS. 40-46) is used. This waler protrudes outward in front of and behind the post alignment. When a similar sized piece of PVC is fixed to the posts, a flat surface is created and use of posts becomes non apparent. This form replaces a rendered masonry wall. A jagged outer surface of the waler aides in the adhesion of the rendering material.

[0039] An exposed waler (FIGS. 47-50) can be used where the PVC waler is to be left exposed as the completed surface. These walers provide a small shadow line to increase aesthetic appeal, per FIG. 48.

[0040] An example procedure for constructing a wall in according to the invention, in this case a tile clad wall using clad walers, is as follows with reference to FIGS. 51 to 70.

[0041] Prior to wall construction, the TOW (Top of Wall) and BOW (Bottom of Wall) levels are established.

[0042] Bore holes are created for the concrete in the ground. If required, the post extensions may be added before post insertion in ground.

[0043] The standard distance between post centres is 2.4m. This can be shortened or lengthened to allow for site specific requirements. The tops of posts are ideally set to height at the TOW, the system allows for a variation of ≤ 20 mm from TOW.

[0044] Once posts are cured, a laser level is used to mark the bottom of the wall on each post. The base plates are fixed with bolted connections at the marked BOW location (FIG. 53) as described above. The remaining post components are installed as per FIG. 54 and described above.

[0045] A base waler (per FIG. 56) is placed on top of the base plates, per FIG. 58. The base waler is a standard shaped waler with minor variation—the vertical holes do not extend completely through the piece, the base of the waler remains solid to retain the waler core fill material, and the SHS is inset4ed in the lower section between the positioning protrusions. The connection protrusions are removed at the lower edges to enable the flat base of the waler to sit flat on the base plate.

[0046] The base waler has a steel square hollow section (SHS) inserted to provide initial rigidity, best seen in FIGS. 32 and 56.

[0047] The base waler is filled with a high slump structural grout to the level indicated by the raised nodule 100 mm from the base of the waler. The grout encapsulates the bolt protrusions through the post and base plate and the SHS within the waler, which when cured provides solid base for subsequent walers.

[0048] Full lengths (6 metres) of reinforcing bar steel are threaded through the holes provided in the posts and overlap subsequent joining pieces. This is continued across the adjoining base waler pieces for the complete span of the wall. The continuous reinforcing will spread the bearing load across multiple posts.

[0049] The oblong shaped openings in the post (seen in FIG. 29) allow for the through placement of reinforcing bar even in the event of some height variation during construction.

[0050] Additional walers are stacked on top of the base waler and connected using the locking mechanism until the required top of wall is reached, as per FIGS. 62-64. Reinforcing steel is added to each subsequent waler level.

[0051] The walers are core filled with a high slump, high strength concrete mix.

[0052] When the wall is ready, cladding panels can be added to the exterior. Pieces of flat plate steel (per FIG. 65) are inserted into the recesses in the waler, per FIGS. 66-68. The pieces are located at intervals to support the cladding material added in the following step.

[0053] Then the cladding is added. The cladding will have a recess running the full length of the cladding pieces (per FIG. 69). The angle of the cladding recess will be the same angle of incidence of the waler recess. The cladding pieces are then positioned onto the steel plate pieces, per FIGS. 69-74. A capping piece is glued to the top surface.

[0054] It will be appreciated by those skilled in the art that the above described embodiment is merely one example of how the inventive concept can be implemented. It will be understood that other embodiments may be conceived that, while differing in their detail, nevertheless fall within the same inventive concept and represent the same invention.

1. A waler adapted for use as part of a modular cast concrete wall system; said waler being adapted to interconnect with adjacent walers; said waler having side walls and upper walls defining an internal casting cavity; wherein said waler is adapted to be oriented in use in a substantially horizontal alignment by having relatively concrete impenetrable side walls and apertures in said upper and lower walls, thereby to permit passage of fluid concrete into and through said casting cavity.

2. The waler of claim 1, incorporating an internal structure adapted to receive, and thereby operably locate, a reinforcing bar passing horizontally through said waler.

3. The waler of claim 2, wherein said internal structure is integrally formed with the waler.

4. The waler of claim 3, wherein said waler has an upper and lower section; said upper section defined by an open top wall narrowing to a V-profile adapted to receive said reinforcing bar; said lower section have in a profile approximating an enclosed rectangle.

5. The waler of claim 4 wherein said waler is constructed from extruded PVC.

6. The waler of claim 1, wherein there are provided rebates in the external wall of said waler that are adapted to allow attachment of external features, such as cladding or the like, to the retaining wall.

7. A modular system for creating a concrete wall, incorporating a plurality of walers according to any preceding claim.

8. The modular system of claim 7, incorporating a base waler that has a lower wall that is relatively concrete impenetrable.

9. The modular system of claim 8, wherein said base waler incorporates a reinforcing insert in said lower section.

10. The modular system of claim 9, wherein said reinforcing insert is a square or rectangular metal tube having apertures in its upper and side walls to allow ingress of concrete.

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